Section X

ACID-BASE DISORDERS

To do common things perfectly is far better worth our endeavor than to do uncommon things respectably.

Harriet Beecher Stowe
(1864)

Chapter 31

Acid-Base Analysis

Seek simplicity, and distrust it.

Alfred North Whitehead (*a*)

Acid-base physiology and pathology is one of the fundamental knowledge bases in the care of acutely ill patients, and mastery is required for the management of critically ill patients. This chapter presents a structured approach to the identification of acid-base disorders based on the relationships between the pH, PCO_2 and bicarbonate (HCO_3) concentration in extracellular fluid (plasma). Much of this information will be a review of what you learned in your general physiology course, but it will now be expanded for use at the bedside. A number of useful reviews are included in the bibliography at the end of the chapter (1–4).

BASIC CONCEPTS

Hydrogen Ion Concentration and pH

The hydrogen ion concentration $[H^+]$ in aqueous solutions is traditionally expressed by the pH, which allegedly is an abbreviation for the *power of hydrogen*, and is a logarithmic function of the $[H^+]$; i.e.,

$$pH = \log (1/[H^+]) = - \log [H^+] \quad (31.1)$$

The physiological range of pH and corresponding $[H^+]$ is shown in Figure 31.1. The normal pH of plasma is 7.40, which corresponds to a $[H^+]$ of 40 nEq/L.

The relationships in Figure 31.1 show that changes in pH are not linearly related to changes in $[H^+]$. Note that the change in $[H^+]$ in the acidotic range (60 nEq/L) is about 2.5 times greater than in the alkalotic range (24 nEq/L), which means that changes in pH will have different implications for acid-base balance at different points along the pH spectrum.

Hydrogen Ion as Trace Element

Also evident in Figure 31.1 is that the concentration of H^+ is expressed in nanoequivalents per liter (nEq/L, or 10^{-9} Eq/L). One nanoequivalent is *one-millionth* of a milliequivalent (1 nEq = 1

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× 10^{-6} mEq), so hydrogen ions are about a million times less dense than the principal ions in extracellular fluid (sodium and chloride), whose concentrations are expressed in mEq/L. This gives hydrogen ions the status of a trace element, but it is a very influential one, as shown by the fact that the normal pH range (7.36–7.44) corresponds to a change in $[H^+]$ of less than 10 nEq/L (!).

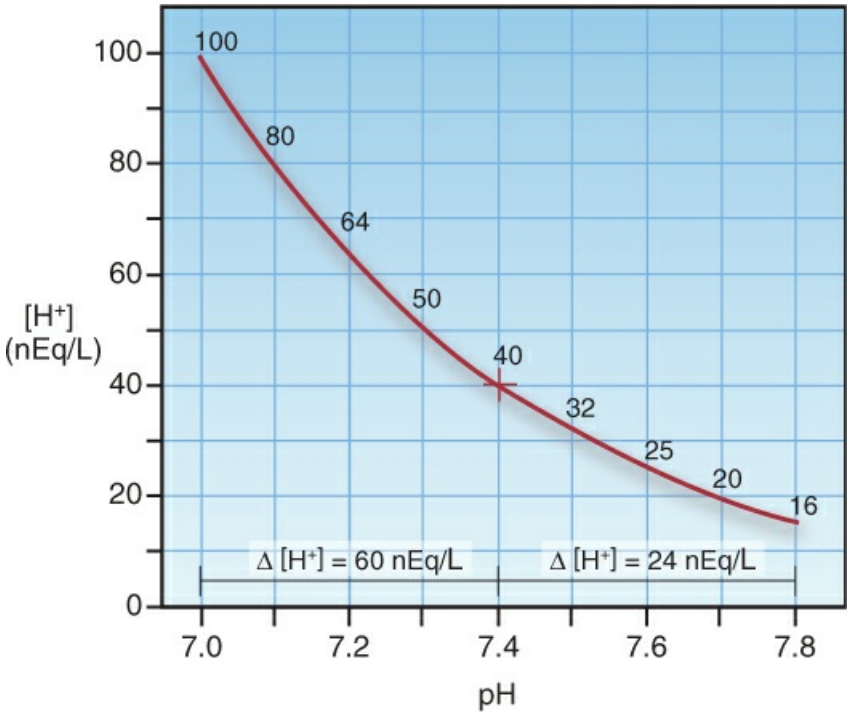


FIGURE 31.1 Graph describing the relationship between the extracellular (plasma) pH and hydrogen ion concentration. Note that the relationship is different in the acidotic and alkalotic ranges of plasma pH.

Classification of Acid-Base Disorders

According to traditional concepts of acid-base physiology, the $[H^+]$ in extracellular fluid is determined by the balance between the partial pressure of carbon dioxide (PCO_2) and the concentration of bicarbonate $[HCO_3]$ in the fluid. This relationship is expressed as follows (3):

$$[H^+] = 24 \times (PCO_2/HCO_3) \quad (31.2)$$

The PCO_2/HCO_3 ratio identifies all the primary acid-base disorders and compensatory responses, as shown in Table 31.1.

TABLE 31.1 Primary Acid-Base Disorders and Compensatory Responses

$\Delta [H^+] \sim \Delta PCO_2/\Delta HCO_3$		
Primary Disorder	**Primary Change**	**Compensatory Response†**
Respiratory Acidosis	↑ PCO_2	↑ HCO_3

Respiratory Alkalosis	↓ PCO_2	↓ HCO_3
Metabolic Acidosis	↓ HCO_3	↓ PCO_2
Metabolic Alkalosis	↑ HCO_3	↑ PCO_2

Primary Acid-Base Disorders

According to Equation 31.2, a change in either the PCO_2 or the HCO_3 will cause a change in the $[H^+]$ of extracellular fluid. When a change in PCO_2 initiates the change in $[H^+]$, the condition is called a *respiratory* acid-base disorder: an increase in PCO_2 is a *respiratory acidosis*, and a decrease in PCO_2 is a *respiratory alkalosis*. When a change in HCO_3 initiates the change in $[H^+]$, the condition is called a *metabolic* acid-base disorder: a decrease in HCO_3 is a *metabolic acidosis*, and an increase in HCO_3 is a *metabolic alkalosis*.

Compensatory Responses

Compensatory responses are designed to limit the change in $[H^+]$ produced by the primary acid-base disorder, and this is accomplished by changing the other component of the $PaCO_2/HCO_3$ ratio in the same direction as the component involved in the primary acid-base disturbance. For example, if the primary change is an increase in $PaCO_2$ (respiratory acidosis), the compensatory response will be an increase in HCO_3, which limits the final change in $[H^+]$. It is important to emphasize that *compensatory responses limit, but do not correct, the change in $[H^+]$ produced by the primary acid-base disorder*. The expected compensatory changes in each of the primary acid-base disorders are shown in Figure 31.2, and each of these is described next.

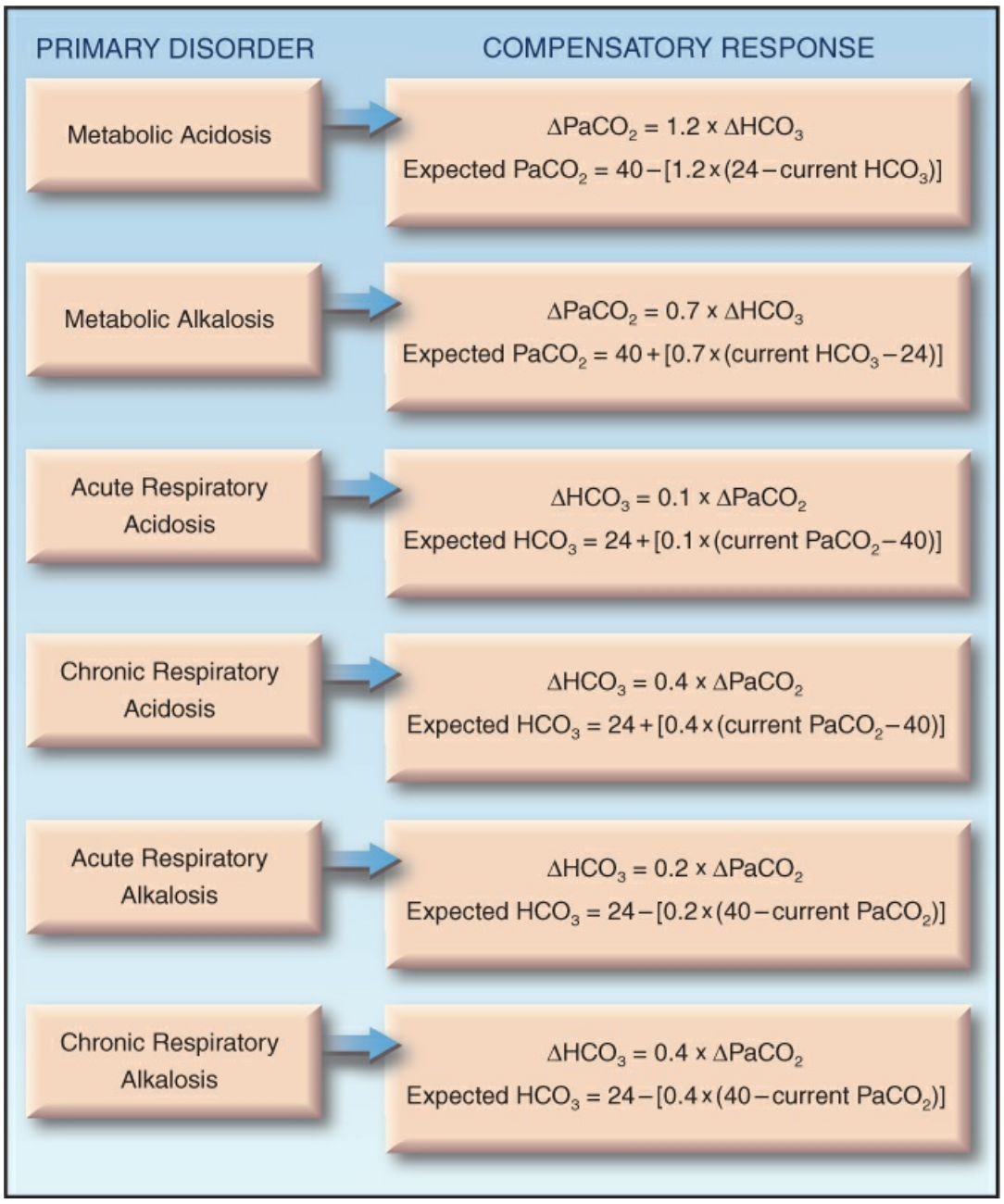


FIGURE 31.2 Predictive equations for evaluating secondary responses to primary acid-base disorders. All equations are from Reference 3.

Responses to Metabolic Acid-Base Disorders

The physiological response to a metabolic acid-base disorder is a change in minute ventilation, which is mediated by peripheral chemoreceptors in the carotid body located at the carotid bifurcation in the neck.

Metabolic Acidosis

The response to metabolic acidosis is an increase in minute ventilation (tidal volume and respiratory rate) and a subsequent decrease in $PaCO_2$. This response appears in 30–120 minutes,

and the magnitude of the response is defined by the equation below (3):

$$\Delta PaCO_2 = 1.2 \times \Delta HCO_3 \quad (31.3)$$

Using a normal $PaCO_2$ of 40 mm Hg and a normal HCO_3 of 24 mEq/L, the above equation can be rewritten as follows:

$$\text{Expected } PaCO_2 = 40 - [1.2 \times (24 - \text{current } HCO_3)] \quad (31.4)$$

EXAMPLE: For a metabolic acidosis with a plasma HCO_3 of 14 mEq/L:

. The ΔHCO_3 is 24 – 14 = 10 mEq/L

. The $\Delta PaCO_2$ is 1.2 × 10 = 12 mm Hg

. The expected $PaCO_2$ is 40 – 12 = 28 mm Hg.

a. If the measured $PaCO_2$ >28 mm Hg, there is a *secondary respiratory acidosis*.
b. If the measured $PaCO_2$ is <28 mm Hg, there is a *secondary respiratory alkalosis*.

Metabolic Alkalosis

The compensatory response to metabolic alkalosis is a decrease in minute ventilation, which increases the $PaCO_2$. This response is not as vigorous as the response to metabolic acidosis because the peripheral chemoreceptors are not very active under normal conditions, so they are more readily turned on than turned off. The expected change in $PaCO_2$ is defined by the equation below (3).

$$\Delta PaCO_2 = 0.7 \times \Delta HCO_3 \quad (31.5)$$

Using a normal $PaCO_2$ of 40 mm Hg and a normal HCO_3 of 24 mEq/L, the above equation can be rewritten as follows:

$$\text{Expected } PaCO_2 = 40 + [0.7 \times (\text{current } HCO_3 - 24)] \quad (31.6)$$

EXAMPLE: For a metabolic alkalosis with a plasma HCO_3 of 40 mEq/L:

. The ΔHCO_3 is 40 – 24 = 16 mEq/L

. The $\Delta PaCO_2$ is 0.7 × 16 = 11 mm Hg

. The expected $PaCO_2$ is 40 + 11 = 51 mm Hg.

a. If the measured $PaCO_2$ is >51 mm Hg, there is a *secondary respiratory acidosis*.
b. If the measured $PaCO_2$ is <51 mm Hg, there is a *secondary respiratory alkalosis*.

Responses to Respiratory Acid-Base Disorders

The compensatory response to primary changes in $PaCO_2$ occurs in the kidneys, where HCO_3

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reabsorption in the proximal tubules changes in the same direction as the change in $PaCO_2$. This renal response is relatively slow, and can take 2 or 3 days to reach completion. Because of this delay, respiratory acid-base disorders are classified as acute or chronic (compensated).

Acute Respiratory Disorders

Acute changes in $PaCO_2$ have a minor effect on the plasma HCO_3, as described by the equations below (3).
For acute respiratory acidosis:

$$\Delta HCO_3 = 0.1 \times \Delta PaCO_2 \quad (31.7)$$

For acute respiratory alkalosis:

$$\Delta HCO_3 = 0.2 \times \Delta PaCO_2 \quad (31.8)$$

EXAMPLE: For an acute change in $PaCO_2$ of 20 mm Hg, the ΔHCO_3 is $0.1 \times 20 = 2$ mEq/L for an acute respiratory acidosis, and $0.2 \times 20 = 4$ mEq/L for an acute respiratory alkalosis. Neither of these changes would be recognized as significant.

Chronic Respiratory Disorders

The renal response to a respiratory acidosis (i.e., an increase in $PaCO_2$) is an increase in HCO_3 reabsorption in the proximal renal tubules, which raises the plasma HCO_3 concentration. The response to a respiratory alkalosis (i.e., a decrease in $PaCO_2$) is a decrease in renal HCO_3 reabsorption, which lowers the plasma HCO_3 concentration. The magnitude of this response is similar, regardless of the directional change in $PaCO_2$, so the equation below applies to both chronic respiratory acidosis and alkalosis.

$$\Delta HCO_3 = 0.4 \times \Delta PaCO_2 \quad (31.9)$$

Using a normal $PaCO_2$ of 40 mm Hg and a normal HCO_3 of 24 mEq/L, the above equation can be rewritten as follows:
For chronic respiratory acidosis:

$$\text{Expected } HCO_3 = 24 + [0.4 \times (\text{current } PaCO_2 - 40)] \quad (31.10)$$

For chronic respiratory alkalosis:

$$\text{Expected } HCO_3 = 24 - [0.4 \times (40 - \text{current } PaCO_2)] \quad (31.11)$$

EXAMPLE: For an increase in $PaCO_2$ to 60 mm Hg that persists for at least a few days:

. The $\Delta PaCO_2$ is $60 - 40 = 20$ mm Hg,

. The ΔHCO_3 is $0.4 \times 20 = 8$ mEq/L,

. The expected HCO_3 is 24 + 8 = 32 mEq/L.

STEPWISE APPROACH TO ACID BASE ANALYSIS

The following is a structured, rule-based approach to the diagnosis of primary, secondary, and mixed acid-base disorders using the relationships between the $[H^+]$, PCO_2, and HCO_3 just described. Several examples are included as instructional aids. The reference ranges for arterial pH, PCO_2 and HCO_3 are shown below.

pH = 7.36–7.44
PCO_2 = 36–44 mm Hg
HCO_3 = 22–26 mEq/L

Stage I: Identify Primary Acid-Base Disorders

The first stage of the approach uses the directional changes in the $PaCO_2$ and pH to identify the primary acid-base disorder.

Rule 1: If the $PaCO_2$ and pH are both abnormal, compare the directional change.

- ***1a:*** If the $PaCO_2$ and pH change in the same direction, there is a primary metabolic acid-base disorder.
- ***1b:*** If the $PaCO_2$ and pH change in opposite directions, there is a primary respiratory acid-base disorder.

EXAMPLE: Consider a case where the arterial pH = 7.23 and the $PaCO_2$ = 23 mm Hg. The pH and $PaCO_2$ are both reduced (indicating a primary metabolic disorder) and the pH is low (indicating an acidosis), so the diagnosis is a *primary metabolic acidosis*.

Rule 2: When the $PaCO_2$ is abnormal but the pH is normal, the condition can be called a *mixed* metabolic and respiratory disorder. The directional change in $PaCO_2$ identifies the type of respiratory disorder, and the metabolic disorder is then in the opposite direction. (This rule is based on the notion that compensatory responses do not correct the primary acid-base change.)

EXAMPLE: Consider a case where the arterial pH = 7.41 and the $PaCO_2$ = 60 mm Hg. Only the $PaCO_2$ is abnormal, so there is a mixed metabolic and respiratory disorder. The $PaCO_2$ is elevated, indicating a respiratory acidosis, so the metabolic disorder must be a metabolic alkalosis.

Stage II: Identify Secondary Acid-Base Disorders

The second stage uses the expected compensatory responses to identify any additional or secondary acid-base disorders. (*Note:* If a mixed acid-base disorder was identified in Stage I, this stage is not necessary).

Rule 3: For a primary metabolic disorder, determine the expected $PaCO_2$.

3a: If the measured $PaCO_2$ is higher than expected, there is a secondary respiratory acidosis.

3b: If the measured $PaCO_2$ is lower than expected, there is a secondary respiratory alkalosis.

EXAMPLE: Consider a case where the arterial pH = 7.28, $PaCO_2$ = 23 mm Hg, and the HCO_3 = 14 mEq/L. The pH and $PaCO_2$ change in the same direction, indicating a primary metabolic disorder, and the pH is below normal, indicating an acidosis, so the disorder is a primary metabolic acidosis. Using Equations 31.3 and 31.4:

. The $\Delta PaCO_2$ is $1.2 \times (24 - 14) = 12$ mm Hg,

. The expected $PaCO_2$ is then $40 - 12 = 28$ mm Hg.

The measured $PaCO_2$ (23 mm Hg) is lower than the expected $PaCO_2$, so there is an additional respiratory alkalosis. Therefore, this condition is a *primary metabolic acidosis with a secondary respiratory alkalosis*.

Rule 4: For a primary respiratory disorder, determine the expected HCO_3 for both acute and chronic (compensated) conditions. If the measured HCO_3 is between the expected HCO_3 for the acute and chronic conditions, it is a *partially compensated respiratory disorder*.

4a: For a primary respiratory acidosis,

1. If the measured HCO_3 is less than the expected HCO_3 for the acute condition, there is a secondary metabolic acidosis.
2. If the measured HCO_3 is higher than expected for the chronic condition, there is a secondary metabolic alkalosis.

4b: For a primary respiratory alkalosis:

1. If the HCO_3 is higher than expected for the acute condition, there is a secondary metabolic alkalosis.
2. If the HCO_3 is lower than expected for the chronic condition, there is a secondary metabolic acidosis.

EXAMPLE: Consider a case where the $PaCO_2$ = 26 mm Hg, the pH = 7.50, and the HCO_3 = 14 mEq/L. The $PaCO_2$ and pH change in opposite directions, indicating a primary respiratory disorder, and the pH is alkaline, so the disorder is a primary respiratory alkalosis. Using Equations 31.8 and 31.9:

. The ΔPCO_2 is $40 - 26 = 14$ mm Hg

. The acute ΔHCO_3 should be $0.2 \times 14 = 3$ mEq/L

. The chronic ΔHCO_3 should be $0.4 \times 14 = 6$ mEq/L

. The expected HCO_3 range is then 18–21 mEq/L.

The measured HCO_3 is 14 mEq/L, which is lower than the expected range of HCO_3 values, so this condition is a *primary respiratory alkalosis with a secondary metabolic acidosis*.

Stage III: Use The "Gaps" to Evaluate a Metabolic Acidosis

The final stage of this approach is for patients with a metabolic acidosis, where the use of measurements called "gaps" can help to uncover the underlying cause of the acidosis. These are described in the next section.

THE GAPS

There are numerous potential sources of a metabolic acidosis in critically ill patients, and the measurements described in this section are designed to help narrow the list of suspects.

The Anion Gap

The anion gap is a rough estimate of the relative abundance of unmeasured anions, and is used to determine if a metabolic acidosis is due to an accumulation of non-volatile acids (e.g., lactic acid) or a primary loss of bicarbonate (e.g., diarrhea) (5,6).

Determinants

To achieve electrochemical balance, the concentration of negatively charged anions must equal the concentration of positively charged cations. This electrochemical balance is expressed in the equation shown below using the most abundant electrolytes in the extracellular fluid, along with the unmeasured cations (UC) and unmeasured anions (UA).

$$(Na) + UC = (CL + HCO_3) + UA \tag{31.12}$$

Rearranging the terms in this equation yields the following relationships:

$$(Na) - (CL + HCO_3) = UA - UC \tag{31.13}$$

The difference (UA − UC) is a measure of the relative abundance of unmeasured anions in extracellular fluid, and is called the *anion gap* (AG). The electrolytes in Equation 31.13 are all routinely obtained in hospitalized patients, and this provides a simple and readily available method for calculating the AG.

$$AG = (Na) - (CL + HCO_3) \tag{31.14}$$

REFERENCE RANGE: The reference range for the AG is typically stated as 8–12 mEq/L (5,6), but electrolyte measurements can differ slightly in different clinical laboratories, so each laboratory should publish their reference range for the AG.

Influence of Albumin

The unmeasured anions and cations that normally contribute to the anion gap are shown in Table 31.2. Note that albumin is the principal unmeasured anion, and thus is the principal determinant

of the anion gap. Albumin is a weak acid that contributes about 2.5 mEq per gram to the anion gap (5,7). A low albumin level in plasma will lower the AG, and this could mask the presence of an unmeasured anion (e.g., lactate) that is contributing to a metabolic acidosis. Since hypoalbuminemia is present in as many as 90% of ICU patients, the following formula for the "corrected AG" (AGc) is recommended (7):

$$\text{Corrected AG} = \text{AG} + [2.5 \times (4.5 - \text{albumin})] \quad (31.15)$$

where 4.5 represents the normal concentration of albumin in plasma (in g/dL).

EXAMPLE: For a patient with a normal AG of 10 mEq/L and a plasma albumin of 2 g/dL, the corrected AG is 10 + (2.5 × 2.5) = 16 mEq/L, so the albumin conversion turned a normal AG into an elevated AG, which has significant implications for determining the cause of a metabolic acidosis, as explained next.

TABLE 31.2 Determinants of the Anion Gap

Unmeasured Anions	Unmeasured Cations
Albumin (15 mEq/L)	Calcium (5 mEq/L)
Organic Acids (5 mEq/L)	Potassium (4.5 mEq/L)
Phosphate (2 mEq/L)	
Sulfate (1 mEq/L)	Magnesium (1.5 mEq/L)
Total UA: (23 mEq/L)	Total UC: (11 mEq/L)

Anion Gap = UA – UC = 12 mEq/L

Using the Anion Gap

The AG can be used to identify the underlying mechanism of a metabolic acidosis, which then helps to identify the underlying clinical condition. An elevated AG occurs when there is an accumulation of fixed or non-volatile acids (e.g., lactic acid), while a normal AG occurs when there is a primary loss of bicarbonate (e.g., diarrhea) (7) . Table 31.3 shows the causes of metabolic acidosis grouped according to the AG.

TABLE 31.3 Causes of Metabolic Acidosis Based on the Anion Gap

High Anion Gap	Normal Anion Gap
L-Lactic Acid	Diarrhea
D-Lactic acid	Isotonic saline infusion
Ketoacids	Early renal insufficiency
Ethylene Glycol (Oxalic Acid)	Renal tubular acidosis
Methanol (Formic Acid)	Acetazolamide

Pyroglutamic Acid†	Ureteroenterostomy
Salicylate Toxicity	
Advanced Renal Failure	

†Also known as 5-oxoproline, and associated with chronic acetaminophen ingestion.

HIGH ANION GAP: Common causes of high AG metabolic acidosis include lactic acidosis, diabetic ketoacidosis, and advanced renal failure (where there is loss of H^+ secretion in the distal tubules of the kidneys). Also included are toxic ingestions of methanol (which produces formic acid), ethylene glycol (which produces oxalic acid), and salicylates (which produce salicylic acid) (5). Two often unrecognized causes of a high gap metabolic acidosis that deserve mention are summarized as follows.

. Lactic acid has two optical isomers (named *dextro* and *levo* by the direction of light reflection), and most clinical laboratories measure only the levo form (L-lactic acid). However, some gut microbes can produce D-lactic acid from carbohydrate metabolism, and a high anion gap metabolic acidosis from elevated D-lactate levels has been reported in patients with short bowel syndrome (8). These patients often develop neurologic manifestations, such as confusion, agitation, ataxia, and nystagmus (9). Although this is an uncommon condition, it will be missed unless a formal request for a D-lactate measurement is submitted to the hospital laboratory. Of interest, D-lactic acid may also contribute to the metabolic acidosis produced by propylene glycol toxicity (5).

. Another often unrecognized cause of a high anion gap metabolic acidosis is 5-oxoproline (also known as pyroglutamic acid), which is associated with chronic ingestion of acetaminophen (10). 5-oxoproline is involved in glutathione synthesis, but why it accumulates in some cases of chronic acetaminophen ingestion is unclear. Serum levels of acetaminophen are usually in the therapeutic range (10). Although this condition is considered rare, the actual incidence is unknown because 5-oxoproline levels are not routinely available.

NORMAL ANION GAP: In the ICU, most cases of normal AG metabolic acidosis are the result of isotonic saline infusions (see Chapter 10), diarrhea, or early renal insufficiency (where there is a defect of bicarbonate reabsorption in the proximal tubules). This type of acidosis is characterized by an increase in chloride concentration (to electrically balance the decrease in HCO_3); hence the term *hyperchloremic metabolic acidosis* is used for normal AG metabolic acidoses. (In high AG metabolic acidoses, the remaining anions from the dissociated acids balance the loss of HCO_3, so there is no associated hyperchloremia.)

The Delta Gap

The delta gap is the ratio of the increase in the anion gap to the decrease in plasma HCO_3, and is described by the following equation:

$$\text{AG Excess}/HCO_3\ \text{Deficit} = (\text{AG} - 12)/(24 - HCO_3) \qquad (31.16)$$

where 12 is the normal anion gap (in mEq/L) and 24 is the normal plasma HCO_3 (in mEq/L). This ratio is sometimes called the *gap-gap* because it involves two gaps (the AG excess and the HCO_3 deficit). The delta gap can be useful in identifying mixed acid-base disorders (11), as

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described next.

Mixed Metabolic Acidoses

In metabolic acidoses caused by non-volatile acids (high AG metabolic acidosis), the increase in the anion gap should be equivalent to the decrease in plasma HCO_3: i.e., the delta gap should be 1.0. (In the real world, differences in distribution and clearance rate create slightly different delta gaps for different acids). However, if there is a second source of acidosis that is a normal AG (hyperchloremic) acidosis, the decrease in HCO_3 is greater than the increase in AG, and the delta gap ratio falls below 1.0. Therefore, *in the presence of a high AG metabolic acidosis, a delta gap <1 indicates the co-existence of a normal AG (hyperchloremic) metabolic acidosis* (12).

DIABETIC KETOACIDOSIS: Diabetic ketoacidosis (DKA) is a high AG metabolic acidosis with delta gap = 1. However, during the management of DKA, the aggressive infusion of isotonic saline creates a normal AG metabolic acidosis. In this situation, the serum bicarbonate decreases relative to the anion gap, resulting in a delta gap <1 (12). Therefore, monitoring the plasma HCO_3 alone will create a false impression that the DKA is not resolving, while the delta gap provides an accurate assessment of the changing acid-base status.

Metabolic Acidosis and Alkalosis

When alkali is added in the presence of a high AG acidosis, the decrease in serum bicarbonate is less than the increase in AG, and the delta gap is >1. Therefore, *in the presence of a high AG metabolic acidosis, a delta-gap >1 indicates the co-existence of a metabolic alkalosis* (4). This is an important consideration because metabolic alkalosis is common in ICU patients (e.g., from the frequent use of diuretics).

Osmolal Gap

The osmolality of plasma can be calculated using the concentrations of the major solutes in extracellular fluid (i.e., sodium, chloride, glucose, and urea): i.e.,

$$\text{Posm} = 2 \times [\text{Na}^+] + \frac{\text{Glucose}}{18} + \frac{\text{BUN}}{2.8} \tag{31.17}$$

where Posm is the plasma osmolality in mosm/kg H_2O, $[Na^+]$ is the plasma sodium concentration in mEq/L, glucose and BUN are the glucose and urea concentrations in plasma in mg/dL, and the factors 18 and 2.8 are the molecular weights of glucose and urea divided by 10, respectively, to express their concentrations in mosm/kg H_2O. The $[Na^+]$ is doubled to include the osmotic activity of chloride.

Using normal plasma concentrations of Na^+ (140 mEq/L), glucose (90 mg/dL), and BUN (14 mg/dL), in Equation 31.17 yields a calculated plasma osmolality of 290 mosm/kg H_2O. The plasma osmolality can also be measured, and the difference between the measured and calculated plasma osmolality is normally <10 mosm/kg H_2O (13). This difference is known as the *osmolal gap*.

Detecting Hidden Solutes

When a solute other than those in Equation 31.17 is added to the extracellular fluid, the measured plasma osmolality will increase, while the calculated osmolality will be unaffected; i.e., the osmolal gap will increase. Thus, an increased osmolal gap can be used to detect an occult toxin (14), and in cases of high AG metabolic acidosis where the common offenders (lactic acid, ketoacids) are not responsible, an elevated osmolal gap can indicate the presence of an occult, unmeasured acid (e.g., D-lactic acid, oxalic acid, formic acid, salicylates, pyroglutamic acid).

A FINAL WORD

Distrusting the Simplicity

For more than 100 years, the evaluation of acid-base balance has been based on a single reaction sequence (shown below), and a single determinant of the plasma pH (i.e., the $PaCO_2/HCO_3$ ratio).

$$CO_2 + H_2O \Leftrightarrow H_2CO_3 \Leftrightarrow H^+ + HCO_3 \quad (31.18)$$

This approach is appealing because of its simplicity, but as Lord Whitehead advises in the introductory quote, simplification can have drawbacks. The following are some major drawbacks with the traditional acid-base paradigm.

. The use of the $PaCO_2$-HCO_3 relationship to identify acid-base disorders is flawed because the CO_2 in plasma is present primarily as HCO_3, so it is difficult to establish an independent identity for HCO_3.

. Bicarbonate does not act as a buffer in the physiological pH range (which is demonstrated in the next chapter). In fact, the plasma $[H^+]$ is a function of the anionic charge equivalence of the plasma proteins, and is not directly related to the plasma HCO_3 (15,16).

There is an alternative (and more valid) physicochemical view of acid-base balance, which was introduced by Peter Stewart (a Canadian physiologist), and the bibliography includes a textbook and original paper by Stewart for any who are interested (15,16).

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Chapter 32

Lactic Acidosis and Ketoacidosis

Fundamental progress has to do with the reinterpretation of basic ideas.

Alfred North Whitehead (*a*)

This chapter focuses on the two most prominent acid-base disorders you will encounter in the ICU: lactic acidosis and ketoacidosis. However, as you will learn, the problem with these disorders has little to do with the acid-base disturbances they promote. Also covered in this chapter is alkali therapy with bicarbonate, and the reasons to avoid it.

LACTIC ACIDOSIS

Lactate is a three-carbon anion that is formed by the chemical reduction of pyruvate in the cytoplasm, in a reaction catalyzed by the lactate dehydrogenase (LDH) enzyme. About 1,500 mmoles of lactate are produced daily under aerobic conditions (1,2), and the principal sites of production are skeletal muscle (25%), skin (25%), red blood cells (20%), brain (20%), and intestine (10%). Activated neutrophils are an additional source of lactate in inflammatory conditions (3). The concentration of lactate in plasma is usually ≤2 mmol/L, with a lactate:pyruvate ratio of 10:1 (1,2). Lactate is cleared from plasma by the liver (60%), kidneys (30%) and heart (10%).

The Evolving Lactate Paradigm

The traditional teaching has been that anaerobic conditions are the principle source of increased lactate production. However, the description of lactate metabolism in Chapter 9 introduces the following notions:

. Lactate production is NOT driven by anaerobic metabolism (see Table 9.5).

. Lactate serves as an oxidative fuel in conditions of metabolic stress, with a caloric yield that is equivalent to glucose (see Table 9.6).

During conditions of metabolic stress, when glucose availability is limited, lactate can

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provide 60% of the energy needs of the myocardium (4), and 30% of the energy needs of the brain (5). During exercise, about three-quarters of the lactate produced is used as an oxidative fuel, and in exercising muscle, there is an intercellular lactate shuttle that transfers lactate from white “glycolytic” muscle fibers to red “oxidative” muscle fibers (6). (Please review the material in Chapter 9 for a more detailed description of these issues.) Another misconception about lactate production that has been known for years, but is rarely mentioned, is described next.

Lactate is Not an Acid

The traditional teaching has been that lactic acid is the end-product of glucose metabolism in the cytoplasm (glycolysis), but there are a number of problems with this scenario, as indicated by the numbers in Figure 32.1 (7,8).

. The end-product of glycolysis is the lactate anion, which is not an acid.

. The chemical reaction that produces lactate actually *consumes* hydrogen ions, which should have the opposite effect to the production of an acid.

. The hydrogen ions associated with lactate production are attributed to the hydrolysis of the 2 ATP molecules formed during glycolysis (8), but how this influences extracellular pH is unclear.

Each of these points is contrary to the traditional explanation for lactic acidosis, and an alternative explanation is warranted. The fact that hyperlactatemia can exist without an associated acidosis (9) is further evidence of the incomplete understanding of the relationship between lactate production and acidosis.

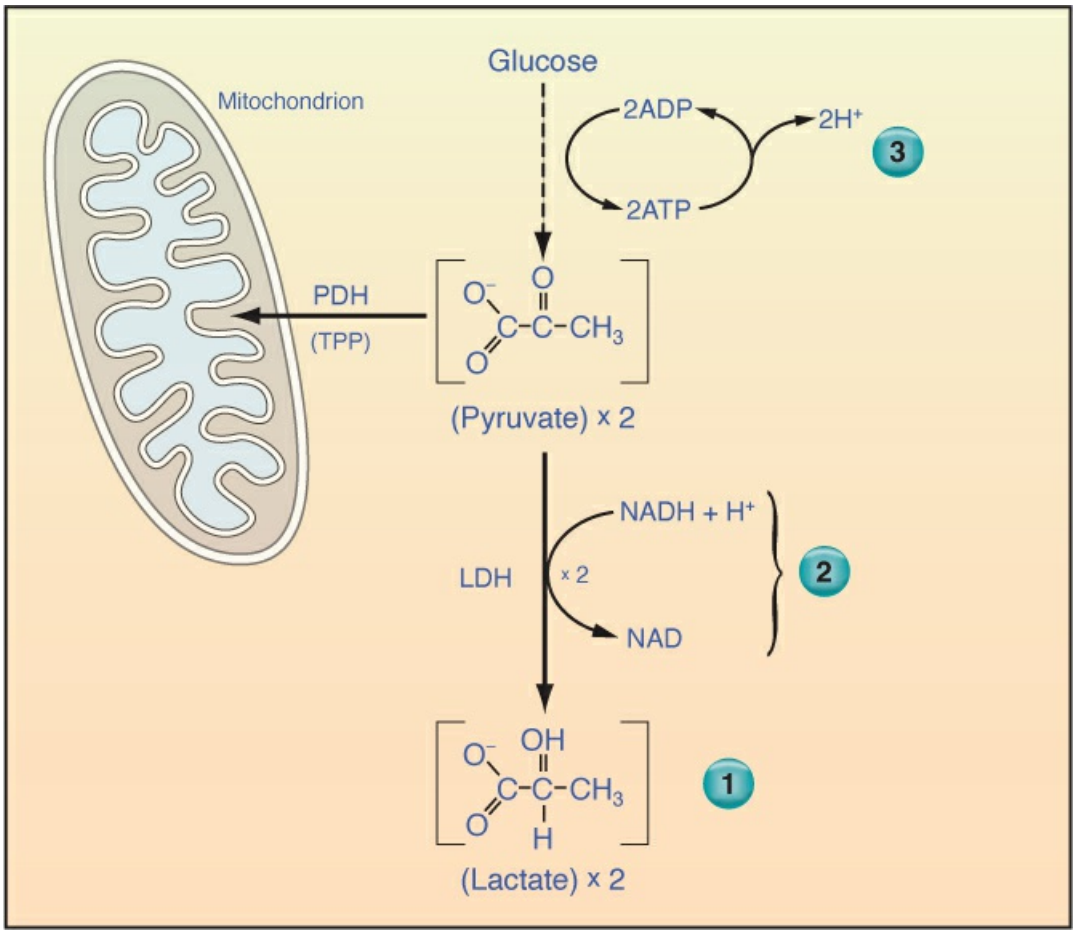


FIGURE 32.1 Problems with the association between lactate production and acidosis. See text for explanation. PDH = pyruvate dehydrogenase, LDH = lactate dehydrogenase, TPP = thiamine pyrophosphate.

Prognostic Value

Regardless of its origins or function, plasma lactate levels have prognostic significance in critically ill patients: i.e.,

. The initial lactate level has a direct relationship with mortality rates, as shown in the left-hand graph in Figure 9.4 (10,11).

. The time required for lactate levels to return to normal (lactate clearance) has greater predictive value than the initial lactate level. Normalization of lactate levels within 24 hours is associated with the lowest mortality rates (see the graph on the right in Figure 9.4) (11).

Causes of Hyperlactatemia

There are numerous causes of elevated plasma lactate levels, and the major ones are listed in Table 32.1.

TABLE 32.1 The Myriad Causes of Hyperlactatemia

Conditions	Drugs and Toxins

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Asthma (acute) Circulatory Shock Compartment Syndromes DKA Liver Failure Leukemia/Lymphoma Mesenteric Ischemia Pheochromocytoma Seizures Sepsis Strenuous Exercise Thiamine Deficiency	Drugs: Antiretrovirals Epinephrine Linezolid Metformin Propofol Toxins: Carbon Monoxide Cyanide Propylene Glycol

Clinical Shock Syndromes

The most notable, and most feared sources of hyperlactatemia are the *clinical shock syndromes;* i.e., hypovolemic, cardiogenic, and septic shock.

SEPTIC SHOCK: Sepsis or septic shock may be the most common cause of lactic acidosis in critically ill patients. The source of elevated lactate levels is a combination of increased production of pyruvate (12) and a defect in O_2 utilization in mitochondria (called c*ytopathic hypoxia*) (13), which may be caused by inhibition of the pyruvate dehydrogenase enzyme responsible for pyruvate entry into mitochondria (see Figure 32.1) (14,15). Tissue oxygenation is not impaired, and can actually be increased (see Figure 17.3).

Systemic Inflammatory Response Syndrome

Systemic inflammation (fever, leukocytosis, etc.) can be accompanied by mild elevations of blood lactate (2 to 5 mEq/L) with a normal lactate:pyruvate ratio and a normal plasma pH. This condition is called *stress hyperlactatemia*, and is the result of an increase in the production of pyruvate without a defect in tissue oxygenation or oxygen utilization (13). The lactate elevation in severe sepsis and septic shock is usually associated with increased lactate:pyruvate ratios and a decrease in plasma pH.

Thiamine Deficiency

The manifestations of thiamine deficiency include high-output heart failure (wet beriberi), Wernicke's encephalopathy, peripheral neuropathy (dry beriberi), and lactic acidosis. The lactic acidosis can be severe (16), and is caused by a deficiency in thiamine pyrophosphate, which serves as a co-factor for pyruvate dehydrogenase (see Figure 32.1). Thiamine deficiency may be more common than suspected in critically ill patients, and should be considered in all cases of unexplained lactic acidosis. (See Chapter 48 for a more detailed description of thiamine deficiency.)

Pharmaceutical Agents

More than 300 medications have been implicated as a cause of elevated plasma lactate levels (17), and the common ones are listed in the right-hand column of Table 32.1.

EPINEPHRINE: Hyperlactatemia is a consequence of high-dose therapy with epinephrine and albuterol (17). Both act by increasing the rate of glycolysis, and produce hyperlactatemia that is aerobic in origin. The albuterol effect may explain reports of lactic acidosis in cases of acute asthma.

METFORMIN: Metformin is an oral hypoglycemic agent that can precipitate lactic acidosis during therapeutic dosing. The mechanism for the lactic acidosis is unclear, but it occurs primarily in patients with renal insufficiency, and has a mortality rate of 17–30% (18,19). Plasma metformin levels are not routinely available, and the diagnosis is based on excluding other causes of lactic acidosis. The preferred treatment is hemodialysis rather than alkali therapy (18,19).

ANTIRETROVIRAL AGENTS: Hyperlactatemia is reported in 8–18% of patients receiving antiretroviral therapy for HIV infection (20). The responsible drugs are the nucleoside analogues (e.g., didanosine, stavudine), and the presumed mechanism is inhibition of mitochondrial DNA polymerase (21). In most cases, the hyperlactatemia is mild and not associated with acidemia, but lactate levels above 10 mmol/L have a reported mortality rate of 33–57% (21).

LINEZOLID: Lactic acidosis is an uncommon but life-threatening complication of therapy with linezolid. The mechanism is unknown, and the reported mortality rate is 25% (22). It is more common in men, and can appear at any time during linezolid therapy.

Non-Pharmaceutical Toxidromes

Lactic acidosis can be the result of intoxications with cyanide, carbon monoxide, and propylene glycol. Cyanide and carbon monoxide intoxication are described in Chapter 53, and the following is a brief description of propylene glycol toxicity.

PROPYLENE GLYCOL: Propylene glycol is used as a solvent to increase the water solubility of some intravenous drug preparations, most notably lorazepam, diazepam, esmolol, nitroglycerin, and phenytoin. About 55–75% of propylene glycol is metabolized by the liver and the primary metabolite is lactate (both levo and dextro isomers) (23). *Propylene glycol toxicity* (i.e., agitation, coma, seizures, hypotension, and lactic acidosis) *has been reported in 19% to 66%* of *patients receiving high-dose intravenous lorazepam for more than 2 days* (24,25). If suspected, the drug infusion should be stopped and another sedative agent (other than diazepam) should be selected. An assay for propylene glycol in blood is available, but the acceptable range has not been determined.

Lactic Alkalosis

Severe alkalosis (respiratory or metabolic) can raise blood lactate levels as a result of increased activity of pH-dependent enzymes in the glycolytic pathway (9,26). When liver function is normal, the liver clears the extra lactate generated during alkalosis, and *lactic alkalosis* becomes evident only when the blood pH is 7.6 or higher. However, in patients with impaired liver function, hyperlactatemia can be seen with less severe degrees of alkalemia.

Other Sources

Other possible causes of hyperlactatemia in ICU patients include *generalized seizures* (from

hypermetabolism) (27), *hepatic insufficiency* (from reduced lactate clearance) (28), *acute asthma* (from enhanced lactate production by albuterol or respiratory muscles) (29), and *hematologic malignancies* (30). Hyperlactatemia that accompanies generalized seizures can be severe (with plasma lactate levels up to 15 mmol/L) but is transient (27).

Diagnostic Considerations

The normal lactate concentration in blood is ≤2 mmol/L, but increases in lactate concentration do not have prognostic value until they reach 4 mmol/L (1). Measurements can be obtained on arterial or venous blood samples. If immediate measurements are unavailable, the blood should be placed on ice to retard lactate production by red blood cells.

The anion gap (described in Chapter 31) should be elevated in lactic acidosis, but there are numerous reports of a normal anion gap in patients with lactic acidosis (31). As a result, *the anion gap should not be used as a screening test for lactic acidosis*.

D-Lactic Acidosis

The lactate produced by mammalian tissues is a levo isomer (l-lactate), whereas a dextro isomer of lactate (d-lactate) is produced by certain strains of bacteria that can populate the bowel (32). D-lactate generated by bacterial fermentation in the bowel can gain access to the systemic circulation and produce a metabolic acidosis, often combined with a metabolic encephalopathy (33). Most cases of d-lactic acidosis have been reported after extensive small bowel resection or after jejunoileal bypass for morbid obesity (32–34).

DIAGNOSIS: D-lactic acidosis can produce an elevated anion gap, but the standard laboratory assay for blood lactate measures only l-lactate. If d-lactic acidosis is suspected, you must specifically request a d-lactate assay.

ALKALI THERAPY

The primary goal of therapy in lactic acidosis is to correct the underlying metabolic abnormality. Alkali therapy aimed at correcting the pH is of questionable value (35). The following is a brief summary of the pertinent issues regarding alkali therapy for lactic acidosis.

Acidosis is Not Harmful

The principal fear from acidosis is the risk of impaired myocardial contractility (36), but this observation comes from isolated heart preparations. *In the intact organism, acidemia is accompanied by an increase in cardiac output* (37), because acidosis stimulates catecholamine release from the adrenal glands.

Protective Role?

Acidosis may have a protective role in life-threatening conditions, since extracellular acidosis has been shown to protect energy depleted cells from cell death (38). The protective role of acidosis is the basis for *pickling*, which is the preservation of food in vinegar (acetic acid), which typically has a pH of 4.0.

Bicarbonate is Not an Effective Buffer

Sodium bicarbonate has limited success in raising the plasma pH in cases of lactic acidosis (39). This can be explained by the titration curve for the carbonic acid bicarbonate buffer system, which is shown in Figure 32.2. The HCO_3 buffer pool is generated by the dissociation of carbonic acid (H_2CO_3):

$$CO_2 + H_2O \Leftrightarrow H_2CO_3 \Leftrightarrow H^+ + HCO_3^- \quad (32.1)$$

The dissociation constant (pK) for carbonic acid (i.e., the pH at which the acid is 50% dissociated) is 6.1, as indicated on the titration curve. Buffers are effective within 1 pH unit on either side of the pK (40), so the effective range of the bicarbonate buffer system is an extracellular pH between 5.1 and 7.1 pH units (indicated by the shaded area on the titration curve). Therefore, *bicarbonate is not an effective buffer in the usual pH range of extracellular fluid*. Bicarbonate is not really a buffer; rather, it is the end-product of CO_2 transport in blood (i.e., the CO_2 added to venous blood enters red blood cells (RBCs), where the carbonic anhydrase enzyme promotes the conversion of CO_2 to H^+ and HCO_3^-. The H^+ is buffered by hemoglobin, and the HCO_3^- is transported out of the RBCs and into plasma in exchange for chloride ions).

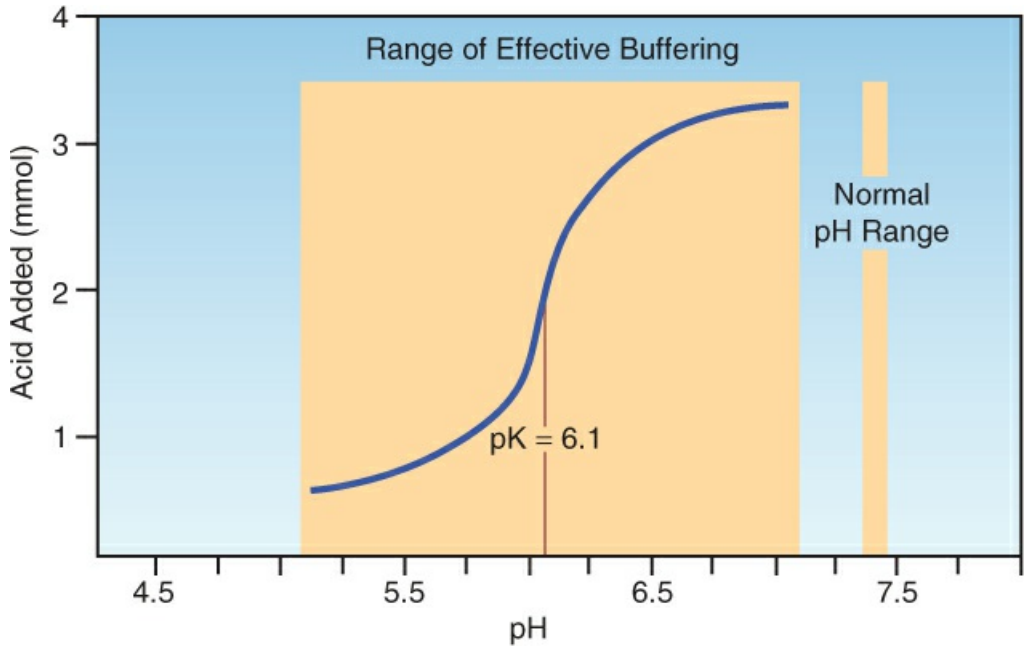


FIGURE 32.2 The titration curve for the carbonic acid-bicarbonate buffer system. The large, shaded area indicates the effective pH range for the bicarbonate buffer system, which does not coincide with the physiological pH range of extracellular fluid. Adapted from Reference 40.

Bicarbonate is Counterproductive

Bicarbonate infusions generate CO_2, which enters cells and cerebrospinal fluid and actually *lowers the intracellular pH* and cerebrospinal fluid pH (41,42). In fact, considering that *the PCO_2 is 200 mm Hg in standard bicarbonate solutions*, bicarbonate is really a CO_2 load that

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must be removed by the lungs.

Bicarbonate infusions can also increase blood lactate levels (42). This effect is attributed to alkalosis-induced increase in lactate production, which is far from desirable for a proposed therapy of lactic acidosis.

Recommendation

Despite the wealth of evidence showing no benefits, and potential harm, from bicarbonate therapy, most (but not all) experts recommend bicarbonate therapy for severe lactic acidosis, when the pH is ≤7.1 and/or the plasma HCO_3 is ≤5 mEq/L) (43,44). It seems more reasonable to include only patients who are hemodynamically unstable (e.g., on vasopressors).

Bicarbonate therapy is probably most helpful in patients with lactic acidosis and renal insufficiency (where part of the problem is loss of HCO_3). The HCO_3 deficit can be estimated as follows (42):

$$HCO_3 \text{ deficit (mEq)} = 0.6 \times \text{wt (kg)} \times (15 - \text{measured } HCO_3) \quad (32.2)$$

(where 15 mEq/L is the desired plasma HCO_3). Half of this deficit can be replaced, and if this results hemodynamic improvement, bicarbonate therapy can be continued to maintain the plasma HCO_3 at 15 mEq/L. If no improvement or further deterioration occurs, further bicarbonate administration is not warranted. The composition of commonly used bicarbonate solutions is shown in Table 32.2.

TABLE 32.2 Composition of Sodium Bicarbonate Solutions

	7.5% $NaHCO_3$	8.4% $NaHCO_3$
Unit Volume (ampule)	50 mL	50 mL
HCO_3 Concentration	0.9 mEq/mL	1 mEq/mL
HCO_3 Content	45 mEq/ampule	50 mEq/ampule
Osmolarity	1,790 mosm/L	2,000 mosm/L

Sodium bicarbonate is also available as 4.2% and 5% solutions, but these are rarely used.

EXAMPLE: For a 70 kg adult with a pH of 7.0 and a plasma HCO_3 of 5 mEq/L, the HCO_3 deficit is $0.6 \times 70 \times 10 = 420$ mEq. One ampule (50 mL) of 8.4% $NaHCO_3$ contains 50 mEq HCO_3 (1 mEq/mL), so 4 amps of 8.4% $NaHCO_3$ (200 mEq HCO_3) would replace about half of the HCO_3 deficit. Since 8.4% $NaHCO_3$ is very hypertonic (2,000 mosm/L), it should be mixed with D_5W and infused slowly (over hours). The ionized calcium should be monitored during bicarbonate infusions, as raising the pH will decrease the ionized calcium, and this can lead to hemodynamic instability. In ventilator-dependent patients, an increase in minute ventilation will also be needed to eliminate the extra CO_2 produced by the bicarbonate infusion.

KETOACIDS

Ketogenesis

When carbohydrates are not available for metabolic energy production, there is a breakdown of triglycerides in adipose tissue to generate fatty acids, which are transported to the liver and metabolized to form three ketones; i.e., acetoacetate, β-hydroxybutyrate, and acetone. This is illustrated in Figure 32.3. These ketones are released from the liver and can be used as oxidative fuels by vital organs such as the heart and central nervous system. The oxidative metabolism of ketones yields 4 kcal/g, which is slightly in excess of the energy yield from the oxidative metabolism of glucose (3.7 kcal/g).

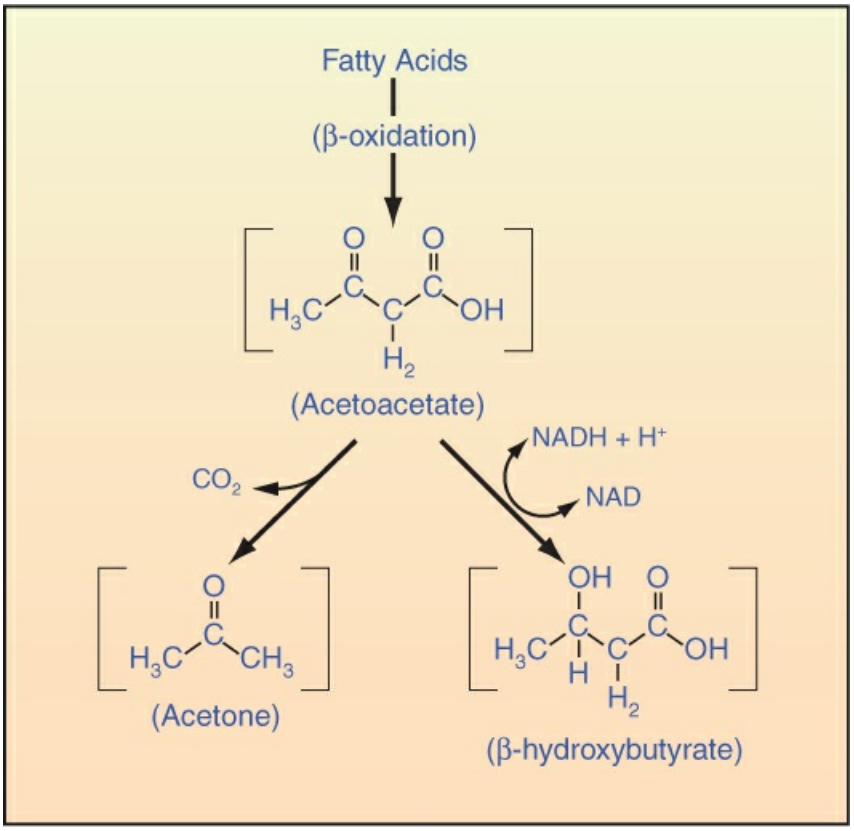


FIGURE 32.3 Ketogenesis in the liver, which occurs in response to diminished availability of glucose. Acetone is a ketone, but is not a ketoacid.

Ketoacids in Blood

The normal concentration of ketones in the blood is negligible (0.1 mmol/L), but blood ketone levels increase tenfold (to 1 mmol/L) after just 3 days of starvation. Acetone is not a ketoacid, but is responsible for the "fruity" odor of breath in patients with ketoacidosis. Acetoacetate (AcAc) and β-hydroxybutyrate (β-OHB) are strong acids, and produce a decrease in plasma pH when their plasma concentrations reach 3 mmol/L (45). The balance of AcAc and β-OHB in blood is determined by the following redox reaction (as shown in Figure 32.3):

$$\text{AcAc} + \text{NADH} + \text{H}^+ \leftrightarrow \beta\text{-OHB} + \text{NAD} \qquad (32.3)$$

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The balance of this reaction normally favors the formation of β-OHB: i.e., in conditions of enhanced ketone production, the β-OHB:AcAc ratio ranges from 3:1 in diabetic ketoacidosis, to as high as 8:1 in alcoholic ketoacidosis. The concentration of ketoacids in the blood in diabetic and alcoholic ketoacidosis is shown in Figure 32.4. Note the preponderance of β-OHB in both conditions.

The Nitroprusside Reaction

The nitroprusside reaction is a colorimetric method for detecting AcAc and acetone in blood and urine. The test can be performed with tablets (Acetest) or reagent strips (Ketostix, Labstix, Multistix). A detectable reaction requires a minimum AcAc concentration of 3 mmol/L. Because *this reaction does not detect the predominant ketoacid, β-hydroxybutyrate* (45), it is an insensitive method for monitoring the severity of ketoacidosis. This is illustrated in Figure 32.4. In alcoholic ketoacidosis, the total concentration of ketoacids in blood is 13 mmol/L, which represents more than a hundredfold increase over the normal concentration of blood ketones, yet the nitroprusside reaction will be negative because the AcAc concentration is below 3 mmol/L.

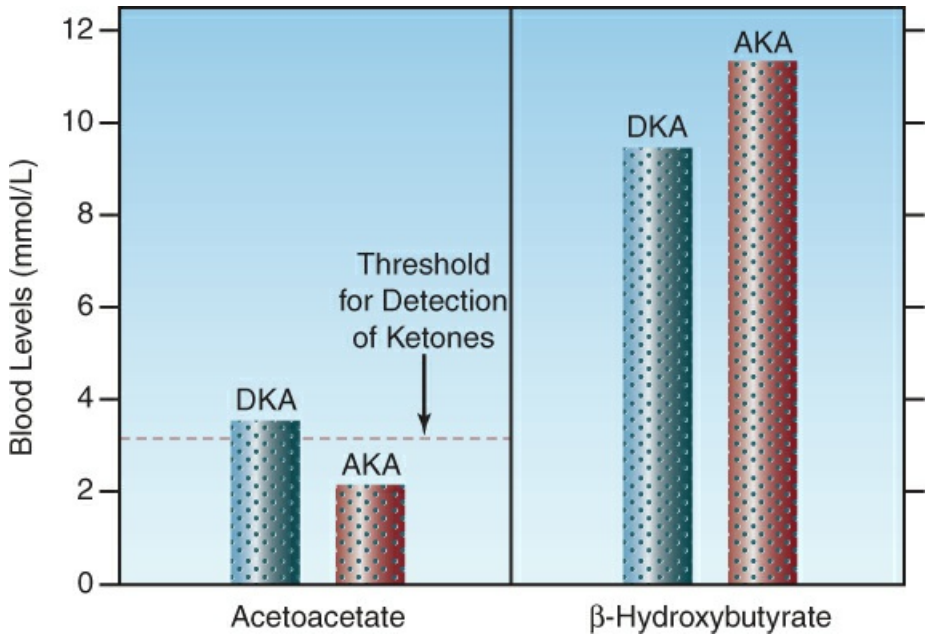


FIGURE 32.4 The concentrations of acetoacetate and β-hydroxybutyrate in the blood in diabetic ketoacidosis (DKA) and alcoholic ketoacidosis (AKA). The horizontal hatched line represents the minimum concentration of acetoacetate required to produce a positive nitroprusside reaction.

β-hydroxybutyrate Testing

There are now portable "ketone meters" that provide reliable measurements of β-OHB concentrations in fingerstick (capillary) blood in about 10 seconds (46). The American Diabetes Association considers measurements of plasma β-OHB with these meters as the preferred method for monitoring patients with diabetic ketoacidosis (47).

DIABETIC KETOACIDOSIS

Diabetic ketoacidosis (DKA) can be seen in patients with type 1 or type 2 diabetes, and there is no previous history of diabetes mellitus in up to 37% of cases (48). The most common precipitating factors in DKA are inappropriate insulin dosing and concurrent illness (e.g., infection). The mortality rate in DKA has decreased to ≤1% (49).

Clinical Features

The features of DKA proposed by the American Diabetes Association include a blood glucose >250 mg/dL, plasma HCO_3 <18 mEq/L, plasma pH ≤7.30, an elevated anion gap, and evidence of ketones in blood or urine (47). However, there are exceptions:

. The blood glucose is <250 mg/dL in about 3% of cases of DKA (49). This condition is known as *euglycemic diabetic ketoacidosis* (50) and it is most common in patients treated with sodium-glucose cotransporter-2 (SGLT2) inhibitors, and in pregnancy and starvation.

. Although uncommon, the anion gap can be normal in DKA (51). The renal excretion of ketones is accompanied by an increase in chloride reabsorption in the renal tubules, and the resulting hyperchloremia limits the increase in the anion gap.

Additional features of interest in DKA are as follows:

. Leukocytosis is not a reliable marker of infection in DKA because ketonemia produces a leukocytosis, which is proportional to the concentration of ketones in plasma (47). However, an increase in immature neutrophils (band forms) can be a reliable marker of infection in patients with DKA (52).

. Elevated troponin I levels without evidence of an acute coronary event has been reported in 27% of patients with DKA (53).

. Dehydration is almost universal in DKA, but this may not be reflected in the plasma sodium concentration because hyperglycemia has a dilutional effect on plasma sodium; i.e., the plasma sodium concentration decreases by 2.4 mEq/L for every 100 mg/dL increase in the plasma glucose concentration (54).

. Hyperglycemia also has a dilutional effect on the plasma chloride concentration, so the measured plasma sodium (not the corrected value) should be used to calculate the anion gap (55).

Management

The management of DKA is summarized in Table 32.3.

Volume Resuscitation

Volume deficits in DKA average 50–100 mL/kg (or 4–8 L for a 175 lb adult) (47). The traditional resuscitation fluid in DKA has been isotonic (“normal”) saline, but balanced salt solutions with a lower chloride concentration (e.g., Plasma-Lyte®, or Ringer’s lactate) have been shown to hasten resolution of the acidosis in DKA (56,57), and these fluids should be preferred to isotonic saline. In cases of euglycemic ketoacidosis, 5% dextrose should be added to all resuscitation fluids.

Volume resuscitation should begin at a rate of 15–20 mL/kg/hr (or 1 L/hour) for the first few

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hours (47). When the patient stabilizes, the infusion rate can be decreased to 250 mL/hr. When the blood glucose falls to 250 mg/dL, change the intravenous fluid to 5% dextrose in 0.45% (half-normal) saline, and decrease the infusion rate to 150–200 mL/hr.

Insulin

Insulin therapy is started only when the plasma K^+ is >3.3 mEq/L (because insulin drives K^+ into cells). Regular insulin is given intravenously, starting with a bolus dose of 0.1 units per kilogram body weight and followed with a continuous infusion at 0.1 units/kg/hr. Because *insulin adsorbs to intravenous tubing*, the bolus insulin dose should be injected directly into the vein, and the initial 10 mL of infusate should be run through the IV setup before the insulin drip is started. The blood glucose levels should decrease by about 100 mg/dL per hour (49), and the insulin infusion should be adjusted to achieve this goal. When the blood glucose level falls to 250 mg/dL, the insulin infusion rate should be decreased to 0.05 units/kg/hr. Thereafter, the blood glucose levels should be maintained between 150–200 mg/dL. Achieving euglycemia is not recommended because of the risk of hypoglycemia.

The insulin infusion is continued until the acidosis resolves (i.e., the anion gap normalizes and the plasma HCO_3 is >18 mEq/L). Subcutaneous insulin can then be started, but the insulin infusion should be continued for at least one hour after the first subcutaneous dose. Patients who were insulin-dependent prior to admission can be placed on their usual outpatient regimen. Patients who are new to insulin should receive 0.5–0.8 Units/kg/day: half of this can be given as a long-acting preparation (e.g., glargine), and the remaining half as a short-acting preparation (e.g., lispro) in divided doses, to cover meals (49).

TABLE 32.3 Management of Diabetic Ketoacidosis

I. Intravenous Fluids

1. Start with a balanced crystalloid fluid (e.g., Ringer's lactate) and infuse at a rate of 15–20 mL/kg/hr (or 1 L/hr) for a few hours.
2. When patient stabilizes, decrease rate to 250 mL/hr.
3. When blood glucose falls to ≤250 mg/dL, change IV fluid to 5% dextrose in 0.45% saline and reduce rate to 150–200 mL/hr.
4. Continue IV fluids until patient can tolerate oral fluids.
5. ***Note:*** In cases of euglycemic ketoacidosis, 5% dextrose should be added to all IV fluids.

II. Insulin

1. Start insulin only if plasma K^+ >3.3 mEq/L.
2. Use regular insulin -start with a bolus of 0.1 U/kg, and inject directly into vein (insulin adsorbs to IV tubing). Then infuse at 0.1 U/kg/hr (run first 10 mL through IV tubing and discard).
3. Adjust rate, if needed, for a decrease in blood glucose of 100 mg/dL/hr.
4. When blood glucose falls to ≤250 mg/dL, decrease rate to 0.05 U/kg/hr.
5. Continue infusion until anion gap normalizes and plasma HCO_3 >18 mEq/L. Then transition to subQ insulin, but continue infusion for at least 1 hr after first subQ dose.

Potassium

Potassium depletion is universal in DKA, and the average deficit is 3–5 mEq/kg (58). However, the initial plasma K^+ is normal or elevated in 96% of patients with DKA because of the

concurrent acidosis (58). The presence of hypokalemia at presentation of DKA indicates severe potassium depletion.

The recommendations for K^+ replacement in DKA are shown in Table 32.4. It is important not to start insulin if the initial plasma K^+ is less than 3.3 mEq/L because insulin drives K^+ into cells, and this can precipitate life-threatening hypokalemia. Regardless of the initial potassium regimen, the plasma K^+ should be checked every 1–2 hours for the first 6 hours.

TABLE 32.4 Potassium Replacement in Diabetic Ketoacidosis

Plasma K^+	Recommendation
<3.3 mEq/L	Hold insulin infusion and give 40 mEq K^+ per hour IV until the plasma K^+ >3.3 mEq/L.
3.3–4.0 mEq/L	Give 20 mEq K^+ per hour IV.
4.0–5.5 mEq/L	Give 10 mEq K^+ per hour IV.
>5.5 mEq/L	Check plasma K^+ every 2 hours.

From Reference 49.

Phosphate

Phosphate depletion is also common in DKA, and averages 1–1.5 mmol/kg (58). However, phosphate replacement has no documented benefit in DKA, and is not recommended unless the phosphate level falls to <1 mg/dL (47,48,58). (See Chapter 38 for recommendations on phosphate repletion.)

Bicarbonate

Bicarbonate therapy has no documented benefit in DKA, and is not recommended as a routine measure (47,48).

Monitoring Acid-Base Status

Normalization of the anion gap is considered evidence that the ketonemia has resolved. The plasma HCO_3 can remain in the acidotic range if isotonic saline is used as the resuscitation fluid, because the high chloride concentration (154 mEq/L) creates a hyperchloremic (normal anion gap) metabolic acidosis that temporarily replaces the high anion gap acidosis from the ketonemia. This resolves when the intravenous infusion is slowed, and should not be a problem if balanced crystalloid fluids are used.

ALCOHOLIC KETOACIDOSIS

Alcoholic ketoacidosis (AKA) is a complex acid base disorder that occurs in chronic, malnourished alcoholics, and typically appears 1 to 3 days after a period of heavy binge drinking (59,60). Several mechanisms are involved, including reduced nutrient intake (which initiates lipolysis and ketogenesis), hepatic oxidation of ethanol (which generates NADH and enhances β-hydroxybutyrate and lactate formation), and dehydration (which impairs ketone excretion in the

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urine).

Clinical Features

Patients with AKA appear debilitated, and typically present with nausea, vomiting, and abdominal pain (59). Electrolyte abnormalities are common, particularly the *hypo's* (e.g., hyponatremia, hypokalemia, hypophosphatemia, hypomagnesemia, hypoglycemia). Mixed acid base disorders are common in AKA, with a high anion gap metabolic acidosis from the ketonemia and mild elevations in plasma lactate, along with a metabolic alkalosis from vomiting. Acidemia is usually mild, and some patients are alkalemic (from protracted vomiting) (60).

Diagnosis

The diagnosis of AKA is suggested by the patient history (i.e., a debilitated alcoholic with a recent bout of binge drinking), along with evidence of dehydration, an elevated anion gap, and the presence of ketones in the blood or urine without hyperglycemia. However, *the nitroprusside reaction for detecting ketones can be negative in AKA*, as shown in Figure 32.4. The oxidation of ethanol in the liver generates NADH, and this favors the conversion of acetoacetate to β-hydroxybutyrate, which results in a low concentration of acetoacetate in blood and urine. Lactate elevations may be present, but are generally mild, and blood ethanol levels can be negative (59). Patients often present with abdominal pain, and pancreatitis should be ruled out.

Management

The management of AKA is notable for its simplicity; i.e., infusion of dextrose-containing saline solutions is usually all that is required. The glucose infusion is needed to correct the acidosis (59,60), while the infused volume corrects volume loss and promotes the renal clearance of ketones. The ketoacidosis usually resolves within 24 hours. Other electrolyte deficiencies are corrected as needed, and thiamine supplementation is recommended because glucose infusions can deplete marginal thiamine reserves (59). (See Chapter 48 for more on thiamine deficiency and replacement.)

A FINAL WORD

Don't Focus on the Acidosis

There are two points in this chapter that deserve emphasis:

. The problem with lactic acidosis and ketoacidosis is not the acidosis, but the underlying condition causing the problem. As a result, the management of these conditions should focus on correcting the underlying cause, and not correcting the acidosis.

. Bicarbonate is not a buffer in the physiological pH range, but is simply an end-product of CO_2 transport in blood. As such, it is an ineffective (and potentially harmful) alkalinizing agent.

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Chapter 33

Metabolic Alkalosis

The first step in the scientific method consists in being curious about the world.

Linus Pauling (*a*)

Although the spotlight usually falls on metabolic acidosis, the most common acid-base disturbance in hospitalized patients is metabolic *alkalosis* (1). The prevalence of metabolic alkalosis can be attributed to three factors: (a) common predisposing conditions, (b) the ability of alkalosis to sustain itself, and (c) the tendency for the condition to go unnoticed and untreated.

This chapter begins with an attempt to simplify the rather complex electrolyte involvement in metabolic alkalosis, and then describes an organized approach to the evaluation and management of metabolic alkalosis. Some comprehensive reviews of the topic are included in the bibliography at the end of the chapter (2,3).

PATHOGENESIS

Metabolic alkalosis is defined as an increase in the bicarbonate (HCO_3) concentration in venous blood to >30 mEq/L (2). The generation and maintenance of this condition involves several factors, including the state of the extracellular volume, the activity of aldosterone, the function of the kidneys, and the plasma chloride and potassium concentrations. Each of these factors is explained briefly in this section.

Bicarbonate Homeostasis

Under normal. steady state conditions, the concentration of bicarbonate in extracellular fluid is kept constant by the actions of the kidneys.

Bicarbonate Reabsorption

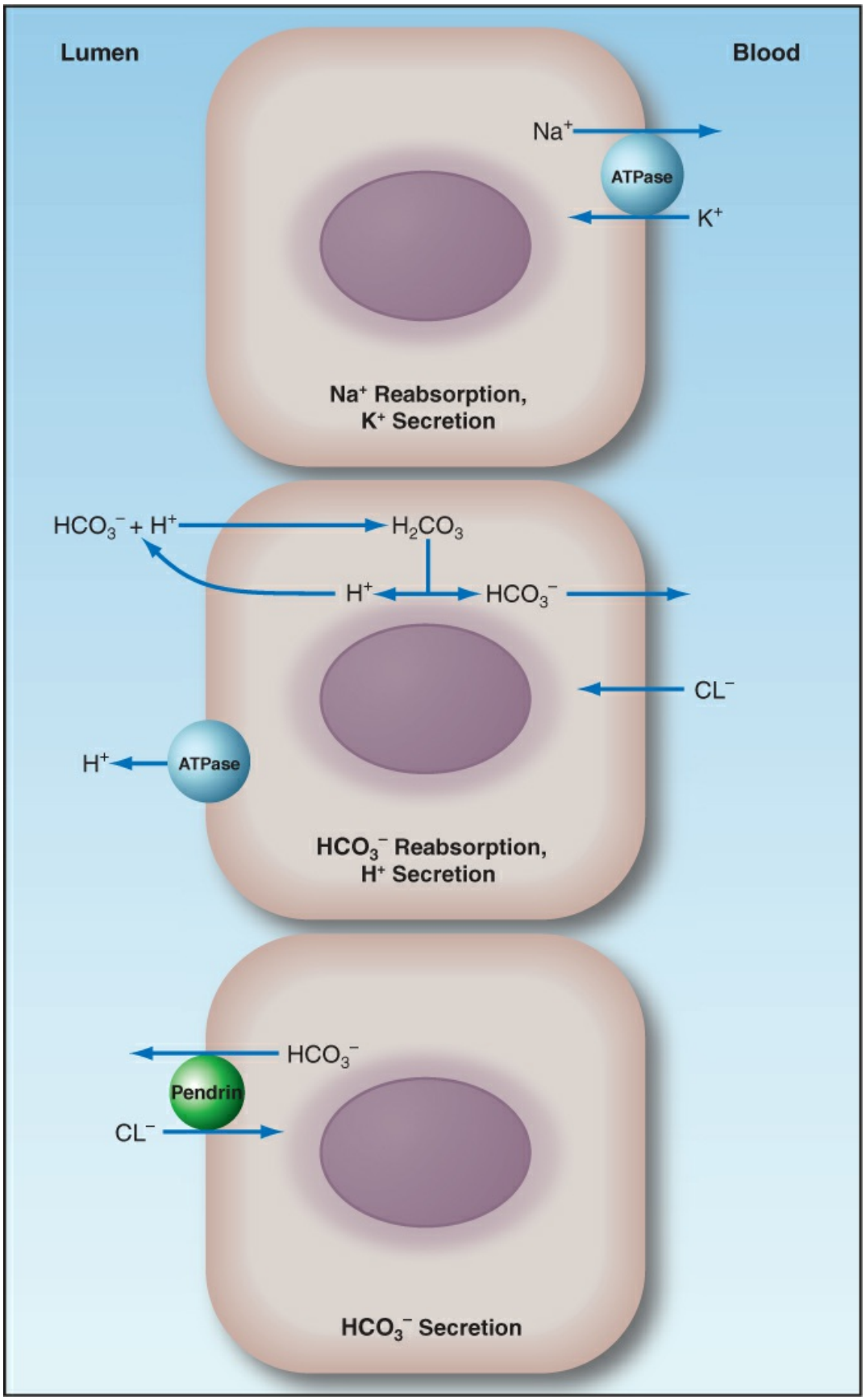
About 4,000–4,500 mEq of bicarbonate (HCO_3) is filtered daily in the kidneys (2), and all of it is reclaimed by the renal tubules. Most (85–90%) of the filtered HCO_3 is reabsorbed in the proximal tubules, which is also the major site for sodium reabsorption. The remaining 10–15%

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of the filtered HCO_3 is reabsorbed by specialized cells in the collecting ducts, and the mechanism for this is shown in the middle cell in Figure 33.1. (Note that chloride moves out of the blood in exchange for HCO_3.) These cells also secrete H^+ into the tubular lumen, which is then excreted as ammonium.

Bicarbonate Secretion

There are other specialized cells in the collecting ducts that are involved in HCO_3 secretion via a chloride-bicarbonate exchange protein (called *pendrin*) that is on the luminal surface of the cells (see the lower cell in Figure 33.1). This process is only minimally active under normal conditions, but the pendrin gene is up-regulated in metabolic alkalosis, to help clear the excess HCO_3 (4).


Lumen
Blood
Na+
ATPase
K+
Na+ Reabsorption,
K+ Secretion
HCO3− + H+
H2CO3
H+
HCO3−
CL−
H+
ATPase
HCO3− Reabsorption,
H+ Secretion
HCO3−
Pendrin
CL−
HCO3− Secretion

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FIGURE 33.1 Specialized cells in the collecting ducts of the kidneys that are involved in acid-base balance. See text for explanation.

Conditions That Promote Metabolic Alkalosis

Several conditions promote the production and maintenance of metabolic alkalosis, as briefly summarized here.

Loss of Gastric Acid

Gastric secretions are rich in hydrogen ions (50–100 mEq/L), chloride (120–160 mEq/L), and to a lesser degree, potassium (10–15 mEq/L) (5). Loss of H^+ in gastric secretions generates an equimolar addition of HCO_3 to the extracellular fluid, which creates a metabolic alkalosis. Loss of chloride and potassium will help to sustain the alkalosis, by mechanisms described later.

Hypovolemia

Many causes of metabolic alkalosis are associated with a decrease in extracellular (plasma) volume. This condition promotes metabolic alkalosis in two ways. First, there is a decrease in glomerular filtration rate, which results in a decrease in filtered HCO_3. Secondly, the decrease in renal perfusion stimulates the renin-angiotensin-aldosterone system, and the aldosterone promotes metabolic alkalosis by mechanisms described next. (*Note:* The metabolic alkalosis associated with hypovolemia is often called "contraction alkalosis", but this is a misnomer because it implies that the problem is a decrease in free water rather than an excess of HCO_3.)

Aldosterone

Aldosterone is a mineralocorticoid produced in the adrenal cortex that stimulates a sodium-potassium exchange pump in the renal collecting ducts (depicted by the uppermost cell in Figure 33.1) to promote Na^+ reabsorption and K^+ secretion. This can lead to K^+ depletion, which promotes metabolic alkalosis by mechanisms described later. Aldosterone also stimulates the membrane ATPase pump responsible for H^+ secretion into the renal tubules (see the middle cell in Figure 33.1) (6).

Chloride Depletion

Chloride depletion plays a major role in promoting metabolic alkalosis by increasing HCO_3 reabsorption and inhibiting HCO_3 secretion in the renal collecting ducts (2,3). Both effects are mediated by a decrease in luminal chloride concentration. The inhibition of HCO_3 secretion is due to inhibition of the anion exchange protein, pendrin, which is the principal mechanism for the ability of chloride depletion to promote metabolic alkalosis (7).

Hypokalemia

Hypokalemia promotes metabolic alkalosis via a transcellular shift of H^+ into cells (in exchange for a shift of K^+ out of cells). The resulting decrease in intracellular pH in renal tubular cells also promotes HCO_3 reabsorption (8).

Diuretics

Thiazide diuretics and "loop" diuretics like furosemide promote metabolic alkalosis via chloride

and K^+ depletion. The principal action of these diuretics is to increase sodium loss in the urine, and urinary chloride excretion usually matches sodium excretion, so this can lead to chloride depletion The increased Na^+ delivery to the distal tubules also promotes K^+ loss via a Na^+–K^+ exchange pump (see the cell at the top in Figure 33.1).

CLINICAL CONSEQUENCES

Metabolic alkalosis is clinically silent in most patients. Severe alkalosis can be accompanied by neurologic manifestations and hypoventilation, but these are rarely life-threatening. In one remarkable case report, an elderly patient with protracted vomiting and a plasma [HCO_3] of 151 mEq/L was hemodynamically stable, and recovered uneventfully after volume and electrolyte replacement (9).

Neurologic Manifestations

The neurologic manifestations attributed to alkalosis include depressed consciousness, generalized seizures, paresthesias, and carpopedal spasms, but these are usually associated with *respiratory alkalosis*, not metabolic alkalosis.

Hypoventilation

The ventilatory response to metabolic alkalosis is hypoventilation, with a subsequent rise in arterial PCO_2. However, this is not a vigorous response, and a considerable rise in plasma HCO_3 may be necessary to produce significant CO_2 retention (10).

The ventilatory response to metabolic alkalosis is described by the following equation (11):

$$\Delta\ PaCO_2 = 0.7 \times \Delta\ HCO_3 \qquad (33.1)$$

where $PaCO_2$ is the arterial PCO_2 and HCO_3 is the plasma (usually venous) HCO_3 concentration. This equation was used to construct the curve in Figure 33.2 showing the relationship between $PaCO_2$ and plasma HCO_3 in progressive metabolic alkalosis. Note that hypercapnia (i.e., $PaCO_2$ > 46 mm Hg) does not occur until the plasma HCO_3 increases almost 10 mEq/L (which represents a 40% increase in plasma HCO_3) and that worrisome hypercapnia (i.e., $PaCO_2$ > 50 mm Hg) does not occur until the metabolic alkalosis is severe.

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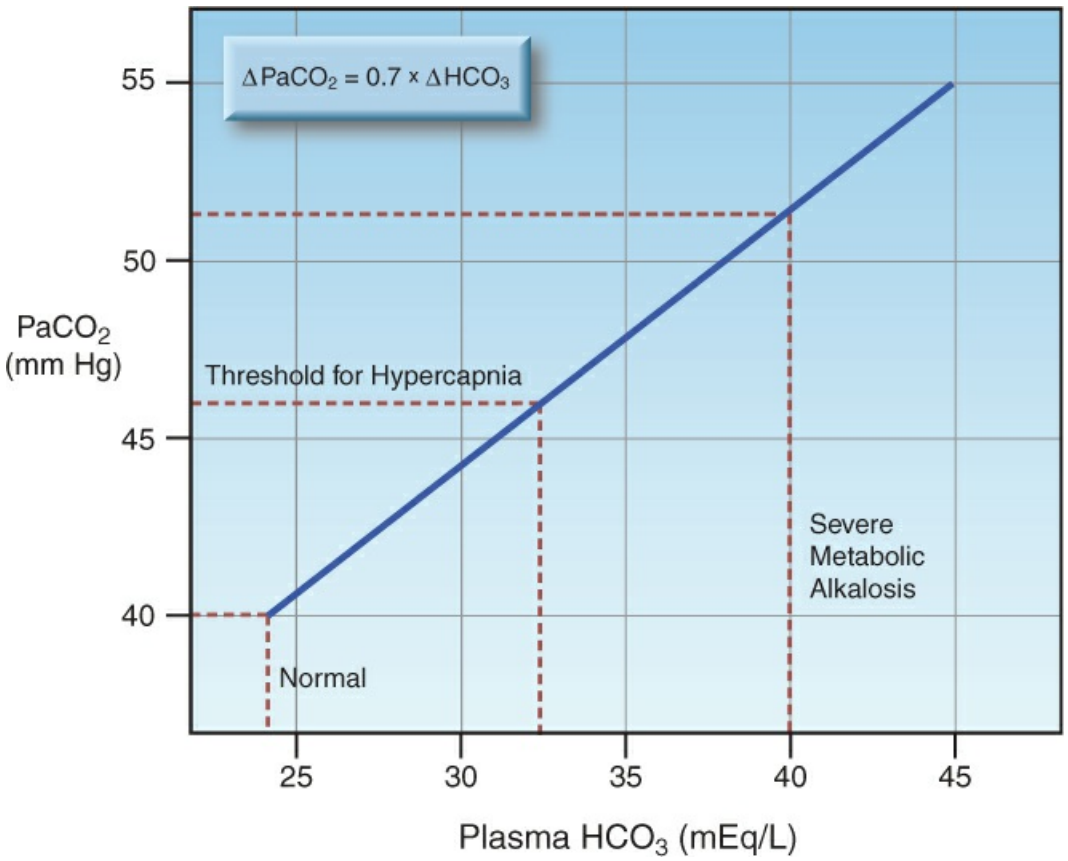


FIGURE 33.2 The relationship between plasma bicarbonate (HCO_3) and arterial PCO_2 ($PaCO_2$) in metabolic alkalosis, as predicted by the equation shown at the top of the graph. Note that the plasma HCO_3 must increase almost 10 mEq/L to produce hypercapnia ($PaCO_2$ > 46 mm Hg), and that worrisome hypercapnia (i.e., $PaCO_2$ > 50 mm Hg) does not occur until the alkalosis is severe.

Oxyhemoglobin Dissociation Curve

Alkalosis shifts the oxyhemoglobin dissociation curve to the left (Bohr effect), which results in a decreased tendency for hemoglobin to release oxygen into the tissues. When the O_2 extraction from capillary blood is constant, a leftward shift of the oxyhemoglobin dissociation curve results in a decrease in venous PO_2 (12), which typically indicates a decrease in tissue PO_2. However, there is no evidence of inadequate tissue oxygenation from this effect.

EVALUATION

The conditions that promote metabolic alkalosis can be organized according to the state of the extracellular volume, and the urinary chloride concentration, as shown in Table 33.1 (2,3).

TABLE 33.1 Principal Sources of Metabolic Alkalosis

Extracellular Volume	Urine Chloride	Conditions
Low	<20 mEq/L	Vomiting, NG Suction

		Thiazide or Loop Diuretics (after-effect) Laxative Abuse
High	>20 mEq/L	Primary Hyperaldosteronism Exogenous Mineralocorticoids Licorice Ingestion
Variable	Variable	K^+ Depletion Mg^{++} Depletion

Low ECV, Low Urine CL

Metabolic alkalosis associated with a low extracellular volume (ECV) and a low urinary chloride concentration (<20 mEq/L) is also called *chloride-sensitive metabolic alkalosis* (2,3). This is the usual type of metabolic alkalosis in ICU patients, and can be the result of gastric acid loss (from vomiting or nasogastric suction), secondary hyperaldosteronism (from hypovolemia), or excessive use of thiazide or loop diuretics (e.g., furosemide). The diuretic effect must be dissipated for the urine chloride to be low.

Laxative Abuse

Diarrhea usually produces a hyperchloremic metabolic acidosis from bicarbonate losses in stool, but the diarrhea associated with chronic laxative abuse is rich in potassium and chloride (70–90 mEq/L), and typically results in a metabolic alkalosis with a low ECV and a low urinary chloride (13). The diagnosis can be elusive, because patients often deny laxative abuse.

High ECV, High Urine CL

Metabolic alkalosis associated with an increased ECV and a high urinary chloride concentration (>20 mEq/L) is also called *chloride-resistant metabolic alkalosis*, and the principal cause is primary hyperaldosteronism. This is an unlikely cause of metabolic alkalosis in the ICU, and is typically the result of aldosterone-producing adrenal adenomas or idiopathic adrenal hyperplasia (14). Analogous conditions include chronic ingestion of licorice, and exogenous mineralocorticoid excess (2,3).

Other Causes

Hypokalemia and hypomagnesemia can cause a metabolic alkalosis in which the extracellular volume status and urinary chloride concentration are variable (3,15).

MANAGEMENT

The management of metabolic alkalosis with a low ECV and a low urinary chloride (i.e., chloride-responsive metabolic alkalosis) involves replenishment of chloride and potassium with isotonic saline and potassium chloride.

Isotonic Saline

The volume of isotonic saline needed to correct a metabolic alkalosis can be estimated by first

estimating the chloride (CL^-) deficit, as shown below (16,17):

$$CL^- \text{ deficit (mEq)} = 0.2 \times \text{wt (kg)} \times (100 - \text{plasma } CL^-) \quad (33.2)$$

where wt is lean body weight in kg, and 100 is the desired plasma chloride concentration (in mEq/L). The required volume of isotonic saline (in liters) is then estimated as follows:

$$\text{Volume of isotonic saline (L)} = CL^- \text{ deficit}/154 \quad (33.3)$$

where 154 is the chloride concentration (in mEq/L) in isotonic saline. This method is summarized in Table 33.2. Rapid saline infusion is not necessary, and a rate of 100 mL/hr above hourly fluid losses is appropriate.

EXAMPLE: A 70 kg adult with protracted vomiting has a metabolic alkalosis with a plasma chloride of 80 mEq/L. The chloride deficit in this case is 0.2 × 70 × (100–80) = 280 mEq. The volume of isotonic saline needed to correct this deficit is 280/154 = 1.8 liters.

TABLE 33.2 Saline Infusions for Metabolic Alkalosis

Step 1. Estimate the chloride deficit:

$$CL^- \text{ deficit (mEq)} = 0.2 \times \text{wt (kg)} \times (100 - \text{plasma } CL^-)$$

Step 2. Determine the replacement volume of isotonic saline:

$$\text{Volume of 0.9\% NaCL (L)} = \frac{CL^- \text{ deficit}}{154}$$

Step 3. Rate of replacement: 100 mL/hr above hourly fluid losses

From References 16 and 17.

Potassium Chloride

Potassium replacement with KCL may be needed in addition to saline infusions. However it important to emphasize that *diuretic-induced hypokalemia can be resistant to potassium replacement if there is concurrent magnesium depletion* (18). Since magnesium depletion is also common during diuretic therapy, the plasma magnesium level should be checked before starting K^+ replacement in patients receiving diuretics. (The evaluation and management of magnesium depletion is described in Chapter 37.)

Edematous States

Patients with edema from heart failure, cirrhosis, or cor pulmonale can have a metabolic

alkalosis from diuretic therapy. In these cases, saline infusion is counterproductive, and K^+ replacement is used to correct the alkalosis. These patients can also benefit from acetazolamide, as described next.

Acetazolamide

Acetazolamide (Diamox) is a carbonic anhydrase inhibitor that inhibits HCO_3 reabsorption in the proximal tubules. The increase in HCO_3 excretion is accompanied by an increase in Na^+ excretion (as $NaHCO_3$), so acetazolamide also acts as a diuretic, which is advantageous in edematous states associated with a metabolic alkalosis. The recommended dose is 250 to 375 mg IV or PO, once or twice daily (19).

Hydrochloric Acid Infusion

Severe metabolic alkalosis that is not corrected with K^+ replacement and acetazolamide can be treated with an infusion of dilute hydrochloric acid, but this can be risky, and is reserved only for patients with severe alkalosis (plasma $HCO_3 > 50$ mEq/L or pH > 7.55).

METHOD: The dose of HCL is determined by estimating the H^+ deficit with the equation below (see also Table 33.3) (19).

$$H^+ \text{ Deficit (mEq)} = 0.5 \times \text{wt (kg)} \times (\text{plasma } HCO_3 - 24) \quad (33.4)$$

where wt is the lean body weight in kg, and 24 is the normal plasma HCO_3.

TABLE 33.3 Hydrochloric Acid Infusions

Step 1. Estimate the hydrogen ion deficit:

$$H^+ \text{ deficit (mEq)} = 0.5 \times \text{wt (kg)} \times (\text{plasma } HCO_3^- - 24)$$

Step 2. Determine the replacement value of HCL:

$$\text{Liters of 0.1N HCL} = \frac{H^+ \text{ deficit}}{100}$$

Step 3. Rate of replacement: ≤0.2 mEq/kg/hr

From References 16 and 17.

The HCL solution is prepared by adding 100 mL of 1 normal (1N) HCL to 900 mL of isotonic saline or sterile water, which makes a 0.1N HCL solution (100 mEq H^+/L) The volume of 0.1N HCL (in liters) needed to correct the H^+ deficit is determined by the ratio (H^+ deficit/100), as shown in Table 33.3. Because HCL solutions are corrosive, they *should be infused through a large, central vein* (20), and *the infusion rate should not exceed 0.2 mEq/kg/hr*

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(17). The entire H^+ deficit does not have to be replaced; i.e., the HCL infusion can be stopped when the plasma pH falls to acceptable levels (e.g., <7.5).

ADVERSE EFFECTS: The major concern with HCL infusions is the corrosive effect on blood vessels. Extravasation of HCL solutions can produce severe tissue necrosis, even when the solution is infused through a central vein (21).

A FINAL WORD

Spotlight on Chloride

Chloride is the second most abundant electrolyte in the extracellular fluid, and has emerged as a major factor in metabolic acid-base disorders, including both metabolic alkalosis (via chloride depletion) and metabolic acidosis (i.e., hyperchloremic metabolic acidosis). It also plays a major role in determining the extracellular volume (which is described in Chapter 35).

In recognition of its importance in acid-base balance and volume regulation, chloride has been given the title "queen of electrolytes" (22), emphasizing that chloride is much more than an inconsequential partner for sodium in the extracellular fluid.

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A L G r a w a n y

Section XI

RENAL & ELECTROLYTE DISORDERS

A man is a bundle of relations.

Ralph Waldo Emerson
(1841)

Chapter 34

Acute Kidney Injury

You can never solve all difficulties at once.

Paul A.M. Dirac (*a*)

As many as 60% of ICU patients experience an acute deterioration in renal function, which has been given the umbrella term, *acute kidney injury* (1). As expected, this condition has a multitude of etiologies, but most of the inciting factors can be classified as inflammatory, ischemic, or nephrotoxic in nature (2,3). The consequences of this condition are as varied as the etiologies, but about 15% of the patients require renal replacement therapy (4), and as many as 60% of the patients will not survive (5).

This chapter begins by presenting the diagnostic criteria for acute kidney injury (including the shortcomings), then proceeds with a description of the most common inciting conditions, and finishes with a description of the renal replacement therapies. Some authoritative reviews of this rather massive topic are included in the bibliography at the end of the chapter (2–4).

DIAGNOSTIC CRITERIA

The term "acute renal failure" was introduced in 1951 (5), but due to a lack of agreement about this condition, the term "acute kidney injury" (AKI) was introduced in 2002, and ten years later (in 2012), the defining criteria for AKI were adopted and published as the "Kidney Disease Improving Global Outcomes (KDIGO) guidelines (6), which are shown in Table 34.1.

TABLE 34.1 Diagnostic Criteria for Acute Kidney Injury

The diagnosis of acute kidney injury (AKI) requires one of the following conditions:

1. An increase in serum creatinine of ≥0.3 mg/dL (≥26.5 μmol/L) within 48 hours.
2. An increase in serum creatinine to ≥1.5 times baseline within the previous 7 days.
3. Urine volume <0.5 mL/kg/hr (ideal body weight) for 6 hours.

From Reference 6.

Limitations

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There are a number of limitations in the diagnostic criteria for AKI, and the major ones are listed below:

. The presence of oliguria (urine output <0.5 mL/kg/hr) can be an appropriate renal adjustment to decreased renal perfusion, rather than a sign of kidney "injury".

. There is a lack of agreement about the minimum increase in serum creatinine required for the diagnosis of acute kidney injury.

. The baseline serum creatinine may not be known.

. The serum creatinine can be misleading as a marker of the glomerular filtration rate, as described next.

Serum Creatinine

The serum creatinine is the traditional marker of the glomerular filtration rate (GFR), and Figure 34.1 shows the popular creatinine-based methods for estimating the GFR. However, the serum creatinine level has the following limitations as a marker of the GFR:

. The creatinine in blood is derived from the creatine produced in muscle, and thus is influenced by muscle mass, and the rate of creatine production in muscle.

. The serum creatinine concentration is influenced by changes in plasma volume.

. Creatinine is not only filtered at the glomerulus, but is also secreted by the renal tubules.

The influence of creatine production in muscle is an important consideration because critically ill patients lose about 2% of their muscle mass each day (7), and a decreased rate of creatine production has been reported in sepsis (8), which is the most common cause of AKI (see later). As a result, the serum creatinine can be an unreliable marker of the GFR in critically ill patients. This is supported by studies showing that *estimates of the GFR based on the serum creatinine level consistently overestimate the true GFR in critically ill patients* (9,10). This has important implications not only for the evaluation of renal function, but also for the appropriate dosing of drugs based on estimates of the GFR.

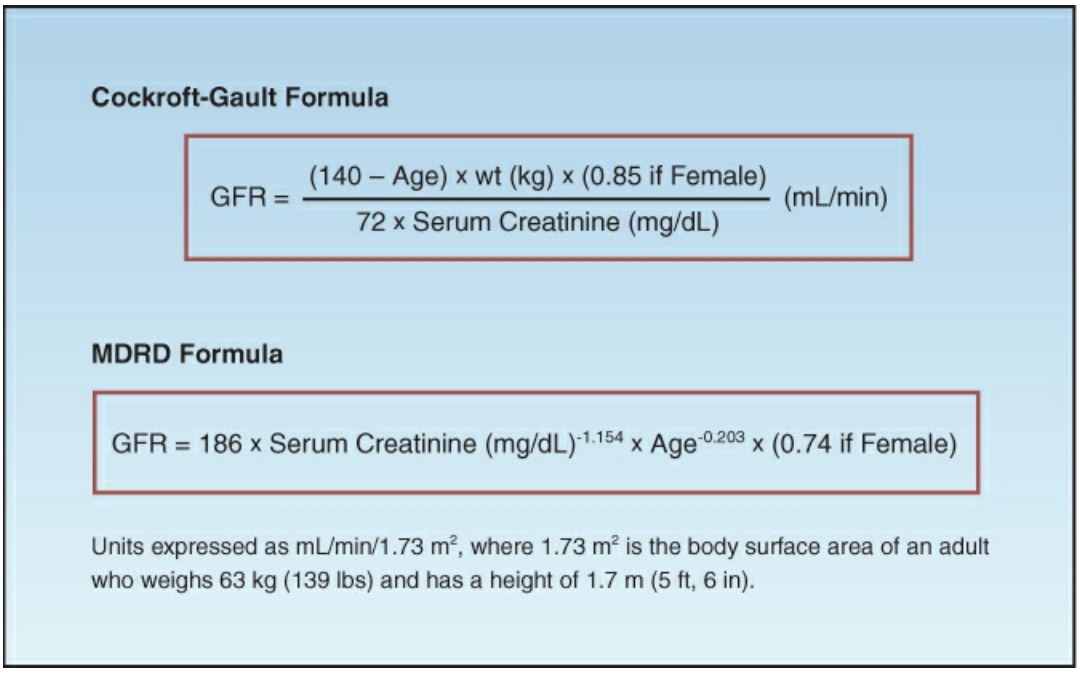


FIGURE 34.1 Popular formulas for estimating the glomerular filtration rate (GFR) based on the serum creatinine level. MDRD = Modification of Diet in Renal Disease.

CYSTATIN C: Cystatin C is a protein produced by most cells in the body that is completely cleared by the GFR. Serum levels of cystatin C are not influenced by muscle mass, which gives cystatin a potential advantage over creatinine. Estimates of the GFR based on the serum cystatin C level are more accurate than creatinine-based estimates, but they often underestimate the true GFR in critically ill patients (9,10). Combining cystatin-based and creatinine-based estimates of GFR increases the accuracy further (9,10), but the complexity of this approach is far from appealing.

Other Monikers

Acute kidney injury that persists for more than a few days is called *acute kidney disease*, and if this condition persists for ≥90 days, it is called *chronic kidney disease* (5). The reasoning for these distinctions is unclear (i.e., injury vs. disease, and 90 days vs. 60 days)

COMMON CAUSES OF ACUTE KIDNEY INJURY

As mentioned in the introduction, most cases of AKI are the result of inflammatory injury, hemodynamic aberrations, or a nephrotoxic agent, and Table 34.2 shows the common causes of AKI in each of these categories.

TABLE 34.2 Common Causes of Acute Kidney Injury

Mechanism	Conditions
Inflammation	Major Surgery Multisystem Trauma Sepsis†

A L G r a w a n y

Hemodynamic Compromise	Cardiac Arrest Heart Failure Hemorrhage Increased Abdominal Pressure Liver Failure
Nephrotoxic	Radiocontrast Dye Rhabdomyolysis Nephrotoxic Drugs & Toxins

†The leading cause of acute kidney injury.

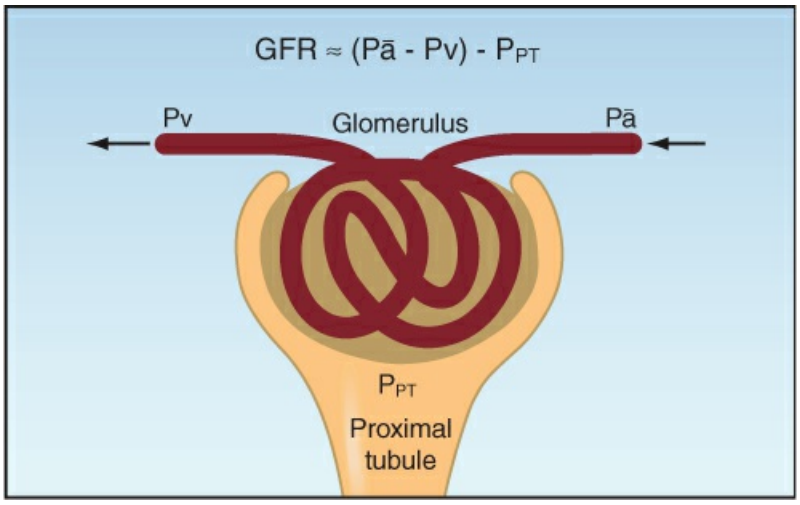


FIGURE 34.2 Schematic depiction of the pressures that influence the glomerular filtration rate (GFR). Pā = mean arterial pressure, Pv = venous pressure, P_{PT} = pressure in the proximal tubules.

Relevant Pressures

Before describing the common causes of AKI, a quick review of the pressures that influence the glomerular filtration rate (GFR) is warranted. This is illustrated in Figure 34.2. Flow through the glomerulus is determined by the difference between the mean arterial pressure and the renal venous pressure (Pā – Pv). This is also called the *glomerular filtration pressure*. The net filtration pressure across the glomerulus, which is also called the *filtration gradient*, is the difference between the glomerular filtration pressure and the pressure in the proximal renal tubules (P_{PT}); i.e.,

$$\text{Filtration Gradient} = (Pā - Pv) - P_{PT} \quad (34.1)$$

These relationships predict that the GFR will decrease not only when there is a decrease in mean arterial pressure, but also when there is an increase in venous pressure (i.e., renal venous pressure or central venous pressure) or an increase in pressure in the proximal tubules.

Sepsis

Sepsis is the leading cause of AKI, and is responsible for about half of the cases of AKI in the

ICU population (3). The principal mechanism is inflammatory (oxidative) injury involving both the capillary endothelium in the glomerulus and the epithelial lining of the renal tubules (11). The renal tubular injury is also called *acute tubular necrosis* (ATN), and is characterized by sloughing of the damaged cells into the lumen of the renal tubules, as shown in Figure 34.3 (12). This creates a luminal obstruction, which increases the pressure in the proximal tubules, and hence decreases the GFR (according to Figure 34.2). This phenomenon is called *tubuloglomerular feedback* (13).

Major surgery and multisystem trauma are other causes of AKI in which inflammation plays a major role (although hemodynamic factors may also be involved). AKI has been reported in over 50% of patients after liver transplantation, in 18% of patients following cardiac surgery, and in 13% of patients undergoing major abdominal surgery (3). Multisystem trauma is accompanied by AKI in about 30% of cases, and rhabdomyolysis is responsible for 30% of these cases (14).

Cardiorenal Syndrome

Heart failure and renal dysfunction are common companions; i.e., in a review that included about 80,000 hospitalized patients with heart failure, renal dysfunction was present in 63% of the patients (15). This association between heart failure and renal impairment is known as the *cardiorenal syndrome* (16).

Venous Congestion

The renal impairment associated with heart failure has traditionally been attributed to a decrease in renal perfusion pressure, but the increase in central venous pressure in heart failure also plays a role in reducing the GFR (17). In fact, in a study of patients with heart failure who were managed with pulmonary artery catheters, the right atrial pressure (central venous pressure) was the only parameter that showed a correlation with renal dysfunction (18). It seems likely that a decrease in renal perfusion pressure plays a major role in the renal dysfunction associated with acute heart failure, while venous congestion assumes an important role in cases of chronic, decompensated heart failure (19).

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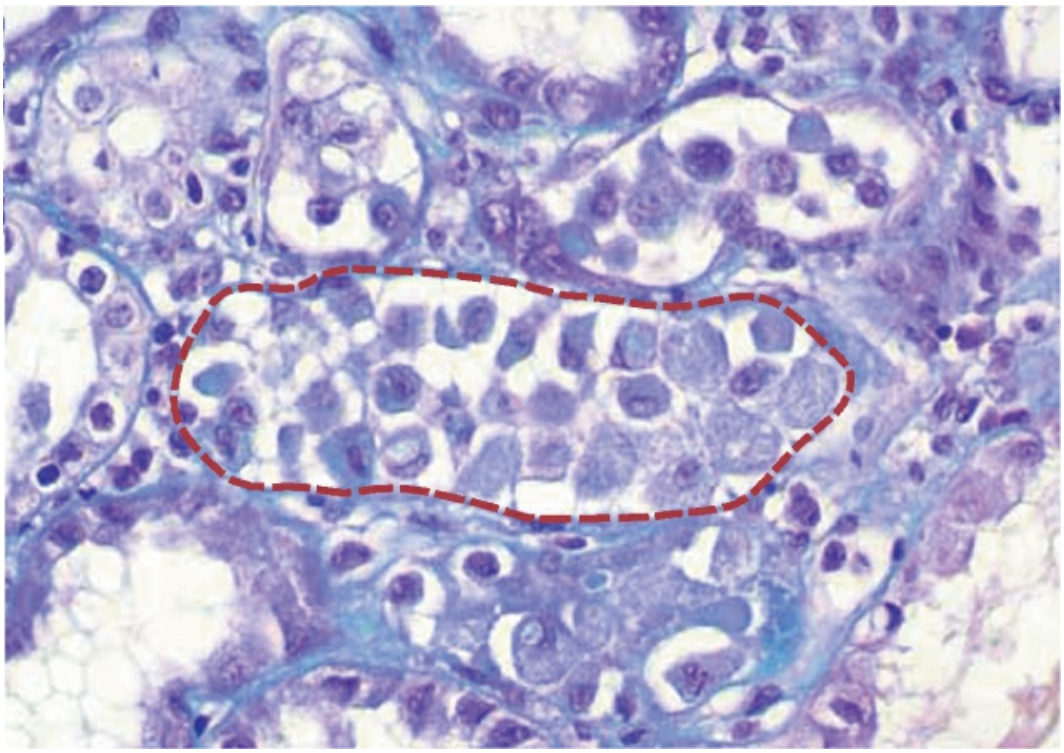


FIGURE 34.3 Photomicrograph of acute tubular necrosis (ATN) showing a proximal tubule (outlined by the dotted line) filled with exfoliated renal tubular cells. Image retouched, from Reference 12.

The influence of venous pressure on the GFR means that AKI can be the result of any disorder that results in pulmonary hypertension and right heart failure. The abdominal compartment syndrome is another cause of AKI mediated by venous pressure, as explained next.

Abdominal Compartment Syndrome

Abdominal compartment syndrome (ACS) is an increase in intra-abdominal pressure that is great enough to cause dysfunction in one or more vital organs (20). The organ dysfunction usually involves the bowel, kidneys, and/or cardiovascular system. *ACS is more common than suspected in critically ill patients, and is often overlooked as a cause of AKI in ICU patients* (21).

Relevant Pressures

The intra-abdominal pressure (IAP) is normally 5–7 mm Hg in the supine position. Intra-abdominal hypertension (IAH) is defined as a supine IAP ≥ 12 mm Hg, and ACS is defined as a supine IAP > 20 mm Hg that is associated with AKI or acute dysfunction in another organ systems.

ABDOMINAL PERFUSION PRESSURE: As shown in Figure 34.2, the driving pressure for glomerular blood flow is the difference between the mean arterial pressure and the pressure in the renal veins (Pā – Pv). However, when IAP exceeds the renal venous pressure, the driving pressure for glomerular flow is the difference between the mean arterial pressure and the IAP. This pressure difference is called the *abdominal perfusion pressure* (APP):

$$APP = P\bar{a} - IAP \quad (34.2)$$

The APP needed to preserve renal blood flow is not well defined, but studies of patients with IAH and ACS have shown that an APP > 60 mm Hg is associated with improved outcomes (20).

FILTRATION GRADIENT: When the IAP is increased, the pressure in the proximal tubules is considered equivalent to the IAP, so the filtration gradient is determined as follows:

$$\text{Filtration Gradient} = P\bar{a} - (IAP \times 2) \quad (34.3)$$

This relationship predicts that an increase in IAP will have a greater impact on the GFR than an equivalent decrease in arterial pressure. This might explain why *oliguria is one of the first signs of ACS* (20).

Predisposing Conditions

ACS is traditionally associated with abdominal trauma or surgery, but several medical conditions can raise the IAP and predispose to ACS, including gastric distension, bowel obstruction, ileus, ascites, bowel wall edema, hepatomegaly, positive-pressure breathing, upright body position, and obesity (22). Several of these factors can co-exist in critically ill patients, which explains why *IAH is discovered in as many as 60% of patients in medical and surgical ICUs* (23).

CRYSTALLOID RESUSCITATION: One of the more common and unrecognized causes of IAH is volume resuscitation with crystalloid fluids, which can raise the IAP by promoting edema in the abdominal organs (especially the bowel). (The distribution of crystalloid fluids is described in Chapter 10: see Figure 10.1.) In one report of ICU patients with a positive fluid balance >5 liters over 24 hours, IAH was present in 85% of the patients, and ACS was diagnosed in 25% of the patients (24). This observation adds to the growing consensus that avoiding a positive fluid balance will reduce morbidity and mortality during the ICU stay. (See Chapter 11 for more on the concerns about fluid accumulation.)

Measuring Intra-abdominal Pressure

The physical examination is insensitive for detecting increases in IAP (25), so IAP must be measured. The standard measure of IAP is the pressure in a decompressed urinary bladder, which is measured with specialized bladder drainage catheters. The following conditions are required for each measurement (20): (*a*) the patient must be in the supine position, with the pressure transducer zeroed in the mid-axillary line, (*b*) a small volume (25 mL) of isotonic saline is injected into the bladder 30–60 seconds prior to each measurement, and (*c*) the IAP is measured only at the end of expiration, and only when there is no evidence of abdominal muscle contractions. The IAP is measured in mm Hg, not cm H_2O (1 mm Hg = 1.36 cm H_2O).

Hepatorenal Syndrome

Acute kidney injury has been reported in 50% of patients with end-stage liver disease (26). The AKI in this setting can be the result of hypovolemia or sepsis (especially spontaneous bacterial peritonitis), or it can be a manifestation of the *hepatorenal syndrome* (HRS), a serious condition caused by the diversion of blood flow away from the kidneys due to the combination of renal

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vasoconstriction and splanchnic vasodilation (27). The diagnosis of HRS is suggested by excluding hypovolemia and sepsis as a cause of AKI, and the urinary indices (described later) can distinguish HRS from ATN.

HRS typically carries a very poor prognosis, with liver transplantation being the definitive treatment. There is, however, a short-term management strategy, which is presented in Chapter 39.

Nephrotoxins

A variety of drugs can precipitate AKI, as shown in Table 34.3, and avoiding these (or any) drugs is always wise when possible (28).

TABLE 34.3 Drugs Most Often Implicated in Acute Kidney Injury

Mechanism	Offending Drugs
Altered Renal Hemodynamics	*Most Cited:* Nonsteroidal anti-inflammatory agents (NSAIDs) *Others:* ACE inhibitors, angiotensin receptor blockers, cyclosporine, tacrolimus
Osmotic Nephropathy	*Most Cited:* Hydroxyethyl starches *Others:* Mannitol, intravenous immunoglobulins
Renal Tubular Injury	*Most Cited:* Aminoglycosides *Others:* Amphotericin B, antiretrovirals, cisplatin
Interstitial Nephritis	*Most Cited:* Antimicrobials (penicillins, cephalosporins, sulfonamides, vancomycin, macrolides) *Others:* Anticonvulsants (phenytoin, valproic acid), H_2 blockers, NSAIDs, proton pump inhibitors

Adapted from Reference 28.

Radiocontrast Dye

Iodinated contrast agents have traditionally been perceived as a cause of AKI in critically ill patients, but this was based on observational studies. More recent and better controlled studies have revealed that *radiocontrast dye injections do not increase the incidence of AKI in critically ill patients* (29), *and they do not worsen renal function in patients with pre-existing AKI* (30). However, iodinated contrast dye is still considered risky in patients with chronic kidney disease and a GFR < 30 mL/min/1.73 m^2 (31).

PROPHYLACTIC FLUIDS: Despite the favorable data on radiocontrast agents, the most recent consensus statement includes the following recommendations (31).

. Prophylactic saline infusions are recommended for radiocontrast dye injections in patients with AKI, or with an estimated GFR < 30 mL/min/1.73 m^2. This recommendation does not necessarily apply to patients with heart failure.

. The infusion should begin 1 hour prior to the study, and should continue for 3–12 hours after the study. There is no firm recommendation regarding the volume infused.

Rhabdomyolysis

AKI is the most serious complication of diffuse muscle injury (rhabdomyolysis). The culprit is myoglobin, which is released by the injured muscle and is capable of damaging the renal tubular epithethial cells (32,33). The source of this injury may be the iron moiety in heme, which is capable of oxidative cell injury via the production of hydroxyl radicals (see Figure 25.6).

DIAGNOSIS: The diagnosis of AKI can be difficult in the setting of rhabdomyolysis because the injured muscle releases creatine, which elevates the serum creatinine level. Elevation of the creatine kinase (CK) in blood is essential for the diagnosis of rhabdomyolysis, but the CK level correlates more with the severity of the muscle injury than the risk of AKI (32). The presence of myoglobinuria is thus a critical component of diagnosing AKI in patients with rhabdomyolysis. A urine myoglobin level >20 μg/L is evidence of significant myoglobinuria (urine usually begins to turn brown at this level) (33).

Myoglobin can also be detected in urine with the orthotoluidine dipstick reaction (Hemastix), which is used to detect occult blood in urine. If the test is positive, the urine should be centrifuged (to separate erythrocytes) and the supernatant should be passed through a micropore filter (to remove hemoglobin). A persistently positive test after these measures is evidence of myoglobin in urine. An alternative approach is to inspect the urine sediment for red blood cells; i.e., a positive dipstick test for blood without red blood cells in the urine sediment can be used as evidence of myoglobinuria.

MANAGEMENT: The following measures are available for preventing or limiting myoglobinuric renal injury (32,33).

. Aggressive volume resuscitation to promote renal tubular flow is the most effective measure for preventing or limiting myoglobin nephrotoxicity. Isotonic saline can be given in volume of 10–20 mL/kg initially, and then infused to maintain a urine output of at least 1 mg/kg/hr. However, if oliguria persists after a few hours, it is important to abandon this strategy to prevent fluid overload.

. Alkalinization of the urine to a pH > 6.5 with bicarbonate infusions can inhibit cast formation in the renal tubules, but this is difficult to accomplish, and bicarbonate has its own problems (see Chapter 32).

. Diuresis with mannitol or furosemide has been used to promote renal tubular flow, but diuresis is risky because it is counterproductive to the planned volume resuscitation.

As many as 30% of patients will require renal replacement therapy (34), which is described later in the chapter.

DIAGNOSTIC CONSIDERATIONS

The following measures can be helpful if there is uncertainty about the diagnosis of AKI or its source.

Ultrasound

A beside ultrasound can be useful for uncovering chronic kidney disease (i.e., small kidneys),

and for identifying post-renal obstruction (i.e., hydronephrosis). This latter condition is especially important, as emergent drainage may be necessary, especially if the urine is infected.

Urinary Indices

The measures in Table 34.4 can help to determine if the source of AKI is renal hypoperfusion (i.e., from hypovolemia or low cardiac output) or renal tubular injury (e.g., from sepsis) (35). However, each of these measures has its limitations. (*Note:* Biomarkers are often mentioned for identifying renal tubular injury, but they are not readily available.)

Spot Urine Sodium

Renal hypoperfusion is accompanied by an increase in sodium reabsorption and a subsequent decrease in urine sodium concentration. In contrast, renal tubular injury results in impaired sodium reabsorption and increased urinary sodium losses. Therefore, when a random urine sample (spot urine) is obtained in a patient with AKI, a urine sodium <20 mEq/L is used as evidence of renal hypoperfusion, while a urine sodium >40 mEq/L is used as evidence of renal tubular injury.

EXCEPTIONS: Unfortunately, there are exceptions to the predicted behavior of the spot urine sodium. For example, renal hypoperfusion can be associated with a high urine sodium (>40 mEq/L) if there is ongoing diuretic therapy, or the patient has chronic renal disease (where there is "obligatory" sodium loss in the urine).

TABLE 34.4 Urinary Indices for the Evaluation of AKI

Measure	Renal Hypoperfusion	Renal Tubular Injury
Spot Urine Sodium	<20 mEq/L	>40 mEq/L
Fractional Excretion of Sodium	<1%	>12%
Fractional Excretion of Urea	<35%	>50%
Urine Osmolality	>500 mOsm/kg	300–400 mOsm/kg
U/P Osmolality	>1.5	1–1.3

Fractional Excretion of Sodium

The fractional excretion of sodium (FENa) is considered a more accurate measure of renal tubular function than the spot urine sodium. The FENa is equivalent to the fractional sodium clearance divided by the fractional creatinine clearance, as expressed by the following equation:

$$\text{FENa}\ (\%) = \frac{\text{U/P}\ [\text{Na}]}{\text{U/P}\ [\text{Cr}]} \tag{34.4}$$

where U/P is the urine-to-plasma ratio for sodium and creatinine concentrations. In euvolemic patients with normal renal function, the FENa is 1% (i.e., only 1% of the filtered sodium is excreted in the urine). When renal hypoperfusion is present, the FENa is <1% (reflecting sodium

conservation), and when renal tubular injury is present, the FENa is typically >2% (reflecting an increase in urinary sodium excretion) (36).

EXCEPTIONS: Like the spot urine sodium, the FENa can be elevated (>1%) by diuretic therapy and chronic renal insufficiency (36). In addition, the FENa can be falsely low (<1%) in patients with acute tubular injury from sepsis (37), or myoglobinuria (38).

Fractional Excretion of Urea

The fractional excretion of urea (FEU) is conceptually similar to the FENa, and is equivalent to the fractional urea clearance divided by the fractional creatinine clearance, as expressed by the following equation:

$$\text{FEU (\%)} = \frac{\text{U/P [Urea]}}{\text{U/P [Cr]}} \qquad (34.5)$$

U/P [Cr]

where U/P is the urine-to-plasma ratio for urea and creatinine concentrations. The FEU is low (<35%) with renal hypoperfusion, and high (>50%) with renal tubular injury. *The FEU is not influenced by diuretics* (39), which is the major advantage over the FENa.

MANAGEMENT

General Considerations

The following comments about the general management of AKI deserve mention.

Fluid Challenges

Volume infusion is typically the initial intervention in patients with AKI, since it is often not possible to rule out renal hypoperfusion as a contributing factor at that time, and the urine output may also be diminished. The fluid challenge is usually 500 mL of a crystalloid fluid infused over 10–15 minutes. (The rate of infusion is more important than the volume). Changes in stroke volume can be used to determine fluid responsiveness, as described in Chapter 11. Fluid challenges should be stopped if there is no response in the stroke volume or urine output.

Diuretics

A trial of intravenous furosemide is reasonable as an attempt to mitigate fluid accumulation in AKI. However, *diuretics should never be given to increase urine output if renal hypoperfusion is a contributing factor in AKI*. Intravenous furosemide does not improve renal function, and is used solely for fluid management (40,41).

Low-Dose Dopamine

Dopamine infusions at low dose rates (2 µg/kg/min) can promote renal vasodilation, which led to the popularity of low-dose dopamine as a treatment for AKI (42). However, this practice fell from favor because it did not improve renal function, and had undesirable effects (e.g., decreased

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splanchnic blood flow) (42,43). The use of low-dose dopamine in AKI is now considered *bad medicine* (from the title of Reference 43).

Renal Replacement Therapy

About 10-15% of ICU patients receive renal replacement therapy (RRT) (4). The usual indications include: (a) symptomatic uremia (e.g., encephalopathy), (b) volume overload, (c) life-threatening, refractory hyperkalemia, and (c) removal of toxin (44).

There is a growing body of RRT techniques, but the descriptions that follow are limited to hemodialysis and hemofiltration. The mechanisms of fluid and solute removal by each of these techniques is shown in Figure 34.4.

Hemodialysis

Hemodialysis is an intermittent form of RRT that removes solutes by *diffusion*, which is driven by the concentration gradient of the solutes across a semipermeable membrane. To maintain this concentration gradient, a technique called *countercurrent exchange* is used, where blood and dialysis fluid are driven in opposite directions across the dialysis membrane. A blood pump is used to move blood in one direction at a rate of 200–300 mL/min, while the dialysis fluid on the other side of the membrane moves at a higher rate (500–800 mL/min) (37).

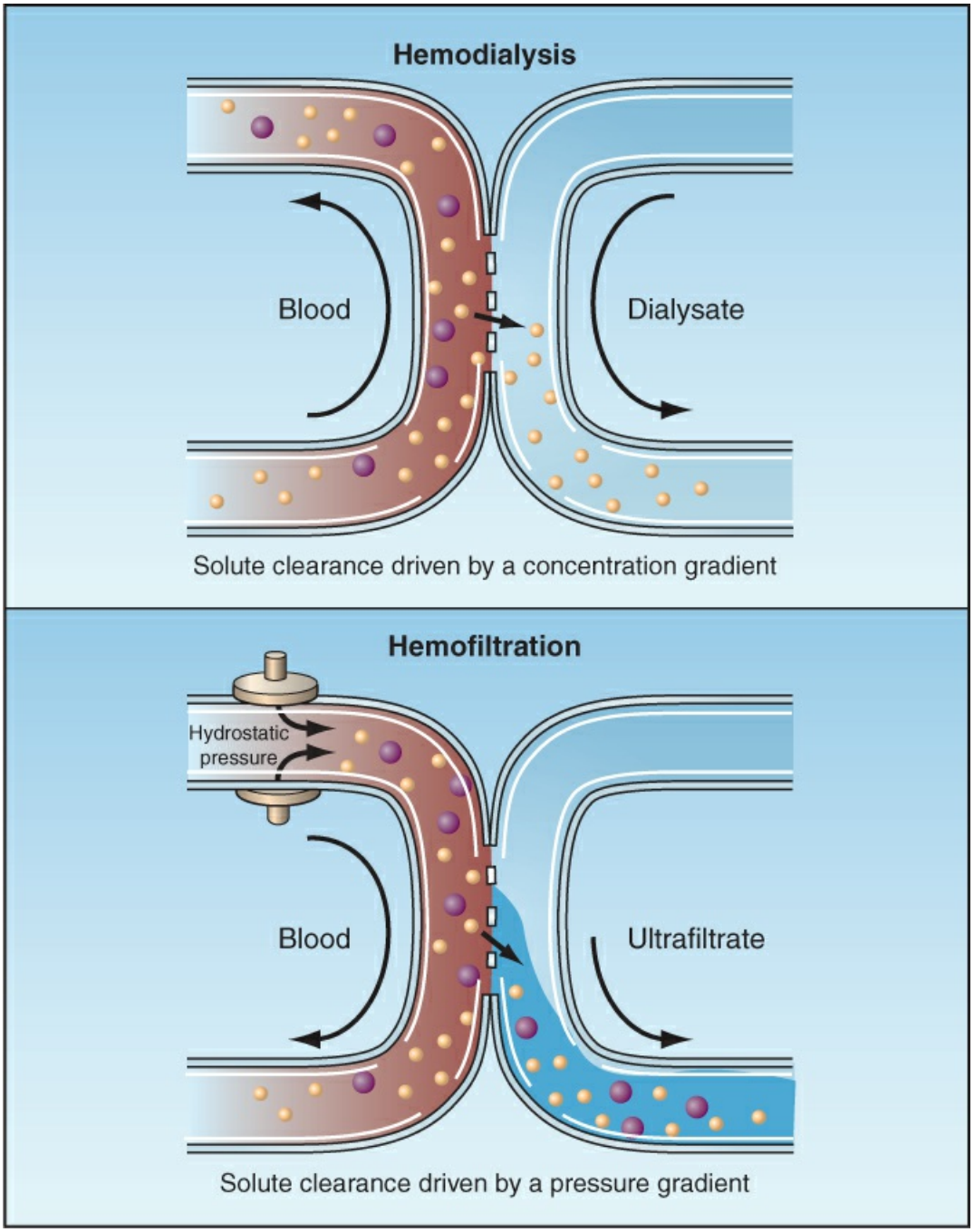


FIGURE 34.4 Mechanisms of solute clearance by hemodialysis and hemofiltration. The smaller particles represent small solutes (e.g., urea), which can be cleared by both techniques, while the larger particles represent larger solutes (e.g., toxins) that can be cleared by hemofiltration, but not by hemodialysis.

Large-bore, central venous catheters like the one in Figure 34.5 are used for acute hemodialysis in the ICU. Note that the catheter has 12-gauge infusion channels, to accommodate the flow rates of 200–300 mL/min used during hemodialysis. These catheters are inserted into the internal jugular or femoral veins in the same manner as other central venous catheters (see Chapter 2). (*Note:* The subclavian vein is not cannulated, which avoids the risk of subclavian

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stenosis that could hamper the selection of access sites for long-term hemodialysis.) Each hemodialysis session lasts about 4 hours, and no more than 2–3 liters of fluid can be removed per session.

ADVANTAGES: The principal benefit of hemodialysis is rapid clearance of small solutes. Only a few hours of hemodialysis is needed to clear a life-threatening accumulation of potassium, or to remove a day's worth of accumulated nitrogenous waste.

DISADVANTAGES: The disadvantages of hemodialysis include: (a) limited removal of fluid and large molecules (drugs and toxins) and (b) the risk of hypotension in patients who are hemodynamically unstable.

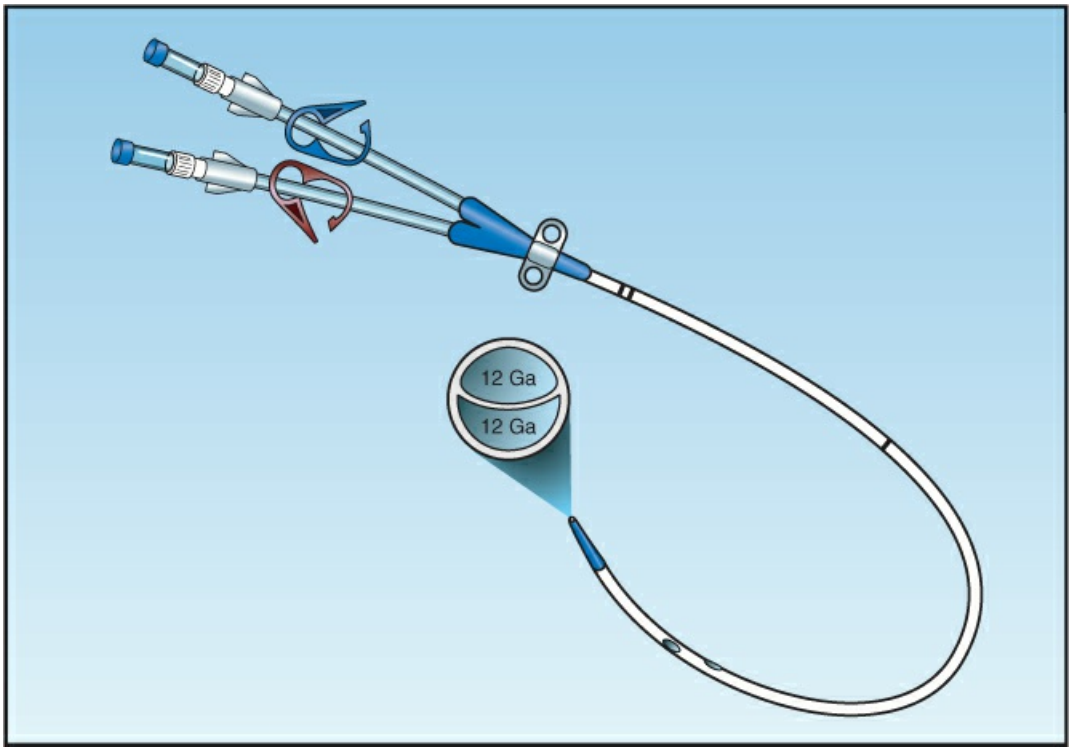


FIGURE 34.5 Central venous catheter for hemodialysis, which has two 12-gauge infusion channels, and is 20 cm (8 inches) in length.

Hemofiltration

Hemofiltration removes solutes by *convection*, where a hydrostatic pressure gradient is used to move a solute-containing fluid across a semipermeable membrane. The bulk movement of fluid "drags" the solute across the membrane, hence this process is also known as *solvent drag* (45).

Hemofiltration can remove large volumes of fluid (up to 3 liters per hour), but the rate of small solute clearance is much slower than during hemodialysis. As a result, *hemofiltration must be performed continuously to provide effective solute clearance*. The popular method is *continuous venovenous hemofiltration* (CVVH), where venous blood is removed and returned through large-bore, double-lumen catheters like the ones used for hemodialysis. Because solutes are cleared with water, the plasma concentration of these solutes does not decrease unless a

solute-free intravenous fluid is infused to replace some of the ultrafiltrate that is lost.

ADVANTAGES: The advantages of CVVH include: (*a*) improved tolerance in hemodynamically unstable patients, (*b*) ability to remove large volumes of fluid, and (*c*) ability to remove drugs and toxins. CVVH with increased pore size is also effective in removing inflammatory cytokines from blood (46), which has important implications for patients with sepsis and septic shock (where the organ dysfunction is inflammatory in origin).

DISADVANTAGES: The major disadvantage of hemofiltration is the slow removal of small solutes, which is not well suited for clearing nitrogenous waste or for life-threatening hyperkalemia.

Hemodiafiltration

Hemodiafiltration combines the benefits of dialysis (rapid clearance of small solutes) and hemofiltration (removal of fluid and toxins), and can be delivered as *continuous venovenous hemodiafiltration* (CVVHD). This has become the favored method of renal replacement therapy in the ICU, although the performance can be fickle.

A FINAL WORD

The Dirac Equation & Acute Kidney Injury

The author of the introductory quote (Paul Dirac) was a theoretical physicist who introduced an equation (the Dirac equation) to describe the behavior of all electrons (*a*). However, the equation ultimately failed in its intended task, which prompted the introductory quote. This has some similarities to the concept of acute kidney injury, which attempts to encompass the entire spectrum of acute renal dysfunction, but ultimately fails in its intended task. (For example, it fails to reliably identify renal dysfunction because it relies on the serum creatinine.) In fact, it is not entirely clear how the term “acute kidney injury” offers anything over the time-honored term “acute renal dysfunction”.

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Cardiorenal Syndrome

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